

Report

(RA)2960
24 January 2025

Report on the budget increases of the MOG II project

Article 23, §2, second paragraph, 2°, of the Law of 29 April 1999 on
the organisation of the electricity market

Non-confidential

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PREAMBLE

Brussels, 24 January 2025

Dear reader,

This document is the result of intensive work by a team of CREG staff and consultants who spent almost three months investigating the causes of an unprecedented budget increase in electricity grid infrastructure. This increase was brought to light by the CREG, which was subsequently tasked by both the executive and the legislative authorities with producing a detailed report on the various figures presented by Elia, in particular on the contracts for the 'direct current part' of the energy island by the end of January 2025.

By the time we finish this report, we hope that a pragmatic solution can be found regarding confidentiality for the hearing of the CREG on this case, planned for 4 February in the Energy, Environment and Climate Committee of the Chamber. Given the social and democratic importance of the case, the CREG has redoubled its efforts, so that its analysis is didactic and understandable for everyone involved in this *a priori* complex case (policymakers, members of parliament, industrial grid users, experts in the sector but also residential consumers and therefore Belgian citizens).

This report presents a wealth of information on the various processes that gradually led to the rise and eventual derailment of costs. These elements allow us to learn from the past and, based on the various recommendations in our conclusion, ensure that the regulator's control and therefore its effectiveness are bolstered, to avoid history repeating itself.

I wish you pleasant reading,

The CREG

EXECUTIVE SUMMARY

Report (RA)2960 on the MOG II project's budget increases is the result of a methodical and objective investigation by the CREG to answer the questions raised by the Minister for Energy and the Chamber of Representatives on the Modular Offshore Grid II (MOG II), in line with the parliamentary hearings on the matter, on 12 and 27 November 2024.

The MOG II project has two objectives. The first objective is to integrate wind energy production from the Princess Elisabeth Zone (PEZ) into the Belgian and European electricity grid. The development of this zone must enable installing 3.15 to 3.5 GW of wind capacity. The second objective of the MOG II is to facilitate access to renewable energy produced elsewhere in the North Sea. This is possible thanks to interconnections such as Nautilus, which links Belgium with the UK.

To fulfil these roles, the design for the MOG II, which was approved by the decision-making federal authorities, consists of the construction of an artificial island for offshore electricity transmission infrastructure. The island needs to be connected to the coast by 2,100 MW of alternating current (AC) high-voltage cables, and 1,400 MW of direct current (DC) cables. The main reason for using DC technology is the necessity to host offshore interconnections which need DC technology because of the long distances to be covered.

Elia divided the MOG II project into five packages that were contracted separately:

- the **artificial island** includes the civil engineering works for the caisson-based construction for the offshore transmission infrastructure;
- the **AC substations** include AC transmission infrastructure. The main elements are 220 kV substations, 66 kV cabins, 220 kV / 66 kV transformers and shunt reactors to compensate for the reactive power. The offshore electrical infrastructure will be installed in steel modules that will be built onshore and then transported to the island and installed on the island, with the exception of transformers and shunts that will be installed directly on the island;
- the **AC cables** consist of six 220 kV AC cables that connect the island to the coast, and run approximately 55 km into the sea;
- the **DC substations** include an offshore AC/DC converter and an onshore AC/DC converter, with a voltage of 525 kV. As is the case for the AC infrastructure, the offshore converter will be built in a module;
- **The 525-kV DC cable system** that connects the offshore AC/DC converter to the onshore AC/DC converter.

The CREG has revealed the significant budget increase for the MOG II project over just three years: according to the latest estimates provided by Elia, this budget has increased from € 2.2 billion in 2021 to € [CONFIDENTIAL: between € 7 and 8] billion at present. Significant cost and/or budget increases can be identified for each of the five packages of the MOG II project. The largest cost increases between the budget from 2021 and the current budget relate to the AC substations (+ 515%), the DC substations (+ [CONFIDENTIAL] %) and the DC cables (+ [CONFIDENTIAL* %]).

If the current design and budget of the MOG II remain as is, and without taking into account the costs of the offshore wind farms and offshore interconnections that need to be connected to the MOG II, the costs for the MOG II that need to be covered by transmission tariffs would amount to around €800 million per year from 2034 onwards, when the complete electricity infrastructure is commissioned. These costs would correspond to €9.8/MWh (or about €34.4/year) for a residential consumer and €4.45/MWh for a large industrial consumer directly connected to the Elia transmission system. After being commissioned, the depreciation and financing costs - which currently represent more than 85% of the costs of the MOG II - will gradually decrease over the next six decades in accordance with the depreciation rate: since the electricity infrastructure of the MOG II will be depreciated over a period of up to 33 years and only the artificial island over a period of 60 years, these costs will decrease faster in the first three decades than in the subsequent three decades.

Over the past three months, the CREG has been investigating the reasons behind this sharp budgetary increase in electricity grid infrastructure. The increase was brought to light by the CREG, who was subsequently tasked by both the executive and the legislative authorities to draft a detailed report by the end of January 2025 on Elia's various figures, in particular on the DC system of the energy island that Elia had not yet contracted.

Consequently, the CREG focused on the DC system of the electricity infrastructure of the energy island in drafting its report. However, the CREG was also able to analyse the AC part of the energy island's electricity infrastructure. Therefore, the conclusions and recommendations in its report go beyond the contracts awarded for the DC system of the electricity infrastructure of the energy island.

The CREG's analyses show that various factors are behind the significant budget increase for the MOG II project:

- the transmission system operator (TSO)'s wish to develop a first-of-a-kind infrastructure, supported by a government decision in the context of the EU's Recovery Plan, the subsidy for which is currently uncertain and limited to €100 million;
- Elia's focus on designing a grid with high availability and operational flexibility, which led to assumptions and design choices that increased the cost base compared to typical offshore transmission designs. These choices included a single AC node design¹ (while Elia plans to operate the MOG II in split AC node), a DC connection with a higher capacity (2,000 MW instead of 1,400 MW), a more expensive bid from the preferred bidder and a high level of redundancy compared to other offshore connection projects currently under development elsewhere in the world. Due to these various technical choices made unilaterally by Elia after 2021, without informing the authorities or taking into account their impact on the electricity consumers' bills, more equipment had to be ordered than originally foreseen in 2021 (and not strictly necessary) to meet the MOG II objectives. In addition to costs that Elia had not originally budgeted for in 2021, these orders for additional equipment also further increased the MOG II budget due to market developments for DC and AC infrastructure over the period 2022-2024;

¹ 'Single node' refers to the electrical connection between the offshore wind turbines, the offshore HVDC converter and the AC export cable.

The electricity infrastructure that makes the connection between the AC and DC parts, and therefore the single node operation, possible was contracted by Elia and will thus be installed on the island. However, initially, the system will be operated in split node mode (i.e. without using the connection between the AC and DC components of MOG II), mainly due to technical and market integration issues that have not yet been resolved. The single node configuration involves significant additional costs and is unusual for offshore connections. If the problems associated with single node operation are not resolved, it is highly likely that the installed electrical equipment will never be used. These elements are described in detail in chapter 5 of the CREG report.

- the market developments for DC and AC infrastructure over the period 2022-2024, due to global scarcity in the supply chain, in particular for offshore applications. These market developments have resulted in limited offers, with fewer and more expensive bids, less flexibility and more risk for Elia;
- keeping to strict timing, whereby unverified assumptions have resulted in premature design choices. The main factors in this regard are: the technical and regulatory feasibility of using a single AC node and the assumption that the offshore DC and AC substations could be built on the island using onshore techniques ('stick-built method'²). In this context, it should be emphasised that if, when the design for the MOG II was decided on in 2021, it had been clear that the stick-built design of the electricity infrastructure on the island was impossible, this would probably have tipped the balance towards the 'with platforms' variant, instead of the chosen 'island' variant.

Based on the detailed analysis carried out by the CREG and an external technical/economic consultant, at least €1.570 billion of the identified cost increase can be directly attributed to unilateral (technical) choices made by Elia^{3,4}. In addition to the above-mentioned quantified elements, a number of possible additional budget increases have been identified that could further (significantly) increase the above-stated amount.

The project under review was recently significantly modified, in particular with regard to the alternatives proposed by the TSO in December 2024.

Based on the alternative options and other information provided by Elia on 13 December 2024 and 8 January 2025, option 3 (Nautilus in point-to-point and 2.1 GW of offshore wind energy in PEZ) seems *a priori* - subject to the working assumptions used in the modelling and subsequent analyses - more favourable from the perspective of Belgian consumers. While this option could significantly reduce the investment costs, it must also be seen in light of the uncertainties currently surrounding the Nautilus project as a whole. Moreover, any possible decision that gives preference to alternative options without a DC system on the island would also mean re-examining the optimal size of the island (currently 6 hectares) and the design of the MOG II's electricity system, which could lead to additional savings for Belgian consumers.

At European level, a general rising trend in investment costs can also be seen in the context of large infrastructure projects, including offshore grid assets. While tensions in the supply chain, higher commodity prices and inflation are the main reasons for the general rise in investment costs, the fact remains that such rises can also be exacerbated by initial underestimates of investment costs by the TSOs in question. To minimise the impact of this rise, or even escalation, of the investment costs, mitigation measures are being considered and discussed by the different countries examined.

Based on the various conclusions in its detailed report and in order to avoid similar budget increases in the future, the CREG has put forward a series of recommendations.

² Stick-built method is a traditional construction method where each component is built on site. All materials are transported to the site and construction is implemented piece by piece.

In 'topside' or modular construction, also known as prefabricated or off-site construction, construction takes place in a factory-controlled environment. The modules are then transported and installed on site (topside) or connected together (modules).

³ For example, the increase in capacity of the DC system from 1.4 to 2 GW and the additional electrical equipment for the single node configuration.

⁴ The amount of €1.570 billion includes the market effects of these choices.

At Belgian level, despite its multidisciplinary expertise, the CREG currently has too limited statutory powers in the context of drafting, approving and possible revising the federal development plan of the transmission system and in the context of the follow-up of infrastructure projects developed by the TSO.

The genesis of the MOG II project, which the CREG describes in its report on the MOG II project's budget increases, illustrates this point.

The CREG was confronted with numerous difficulties in the context of monitoring the evolution of costs related to the MOG II project. These difficulties included:

- the TSO's lack of transparency regarding all the information which is useful for the proper exercise of this monitoring;
- the lack of quantitative and qualitative analyses that are essential before the federal government approved the MOG II design in 2021;
- the TSO's refusal to analyse alternative options for the MOG II design at the request of the CREG in June 2024, to which no positive follow-up was given until December 2024, in order to assess the alternatives and make informed investment choices in the public interest and in the interest of Belgian consumers;
- the fact that until October 2024, and despite the explicit request of Elia's board of directors in March 2024, the TSO did not proactively and timely inform the competent authorities of the significant developments in the project. These include the identified budget overrun and the TSO's unilateral decisions, in particular on significant changes regarding the choice of technical options that led to the budget increases. The CREG would however like to highlight that the cooperation with the TSO proved constructive in the context of the drafting of this CREG report on the MOG II project's budget increases after October 2024, and it recommends remaining open and transparent towards the CREG.

The recommendations below aim to improve the regulatory framework for investment projects in grid infrastructure. They pertain both to the MOG II project examined here and to other ongoing or future investment projects.

As they address issues of public interest, it would appear to be crucial first and foremost that grid infrastructure projects are subject to legal procedures that guarantee, among other things, the objectivity and transparency of all qualitative and quantitative analyses underpinning them. This requirement is essential to ensure that the results of these analyses only serve the public interest, without being influenced by possible conflicts of interest or other specific considerations.

To ensure that the public interest remains the absolute priority in guiding the decisive investment choices for Belgium's energy future, the Electricity Act should also be amended, with regard to:

- the current procedure for preparing, approving and following-up grid development plans, in particular by:
 - strengthening the role of the public authorities, including the CREG;
 - obliging the TSO to present, in the very first phases of elaborating the development plan, all possible investment options, supported by quantified analyses to meet the same grid needs, so that the authorities have a certain decision-making flexibility. Next, the TSO should adequately describe the conceptual design in the development plan. Finally, the TSO should avoid putting the competent and decision-making authorities in a bind, where alternative choices and possibilities are limited or non-existent, to the detriment of the public interest;
 - a strict budgetary monitoring mechanism for the TSO's investment projects, including periodic reports to the competent and decision-making authorities. This would describe the evolution of costs compared to the initial estimates, as well as the explanatory factors behind any identified discrepancies, in order to detect any budget overruns or even budget slippages and consider corrective measures in good time;
- the tariff framework, to ensure better control of the investment costs and promote proactive management by the TSO in the interest of grid users. This could be through a new provision in the tariff guidelines to offer a real incentive for strict budgetary monitoring;
- the restrictions relating to professional secrecy currently imposed on the CREG, to allow genuine parliamentary debate and democratic follow-up on issues of public interest and services provided by the CREG, in particular at the request of the public authorities. Providing a detailed report is indeed crucial to ensure the efficient functioning of the regulator and the parliamentary debate on the matter.

In addition to these considerations, there is also a need to take into account the possible views of different stakeholders as much as possible throughout the process, based on public consultations, so as to consolidate public support for decisive investment projects in the context of energy transition.

INTRODUCTION

This report is the result of a systematic and objective investigation by the CREG to answer the questions raised by both the executive (§ 5, chapter 2 Antecedents) and the legislative authorities (§ 10, chapter 2 Antecedents) regarding the MOG II.

The CREG solicited DNV for this investigation, a technical-economic consultant selected for its advanced expertise, especially in the field of HVDC systems.

The report is divided into nine chapters, including:

- To provide context:
 - an explanation of the existing legal situation (chapter 1 Legal framework: powers of the CREG)
 - an objective analysis of the background to the case (chapter 3 Background to the MOG II project), with justification documents, the process of the various decisions taken which are not based on the CREG's expertise, or only to a limited extent;
 - the passing on of the cost of the MOG II in the bills of electricity consumers (chapter 4 Estimate of the MOG II impact on electricity bills);
- To provide detailed analysis:
 - the technical-economic analysis (chapter 5 Detailed analysis of the investment for MOG II);
 - the comparative cost-benefit analysis (chapter 6 Comparison of costs and benefits of the possible variants for MOG II);
 - feedback from European regulators (chapter 7 European regulators).
- Finally, based on the various findings, the report offers recommendations for amending the legal framework (chapter 8) to strengthen the role of the regulator. Furthermore, the report summarises the different assessments into a general conclusion (chapter 9) to prevent similar situations to the MOG II case from recurring in the future.

Various external analyses and position papers were also considered as a contribution to the discussion and as an additional source of inspiration. In particular, in line with its strategic plan 2024-2029 and its memorandum 2024, the CREG emphasises again the urgent need to be permanently conscious of keeping system costs at an affordable level, in the general interest and in the interest of Belgian consumers. In this context, and to paraphrase Febeliec:

"We don't know whether an efficient energy transition will be affordable. We do know that an inefficient transition will not be affordable. Focus on efficiency. Go for the lowest total system cost."

In this context, the CREG has a key role to play, taking into account the legal powers entrusted to it.

This report was approved at first reading by the Board of Directors on 24 January 2025 and sent to Elia so they could identify any material errors on the one hand, and any commercially sensitive information in the report on the other (see attached letter).

The first version of the report is confidential. In accordance with Article 26, § 2 of the Electricity Act, the members of the CREG are subject to professional secrecy and, as such, may not disclose commercially sensitive or personal data of which they have become aware in the course of their duties, apart from exceptional cases. Only the non-confidential version, drawn up after consultation with Elia, may be communicated to third parties, subject to the exceptions provided for by law, and/or published.

Considering the Parliament's request, the preliminary non-confidential report will be submitted for consultation to interested parties in the case.

1. LEGAL FRAMEWORK: POWERS OF THE CREG

1. The MOG II is included in the 2024-2034 federal development plan of the electricity transmission network, approved by ministerial decree of 5 May 2023. In this context, based on the law of 29 April 1999 on the organisation of the electricity market (Electricity Act) and the Royal Decree of 12 May 2024 laying down the procedure for the preparation, approval and publication of the grid development plan⁵, the CREG has very limited powers of action, both *a priori*, when the federal development plan is drawn up, and *a posteriori*, to amend it after approval by the minister.

Prior to the approval of the federal development plan by ministerial decree, the CREG's power is limited to issuing a simple, non-binding opinion (Articles 6/4, § 2 and 13, § 1 of the Electricity Act). Moreover, if the CREG does not respond to the request for an opinion within 40 days, its opinion is deemed favourable⁶. This situation, about which the CREG has repeatedly warned the government⁷, including in its memoranda⁸, highlights the lack of effective powers and mechanisms to have an influence before the minister's approval of the federal development plan.

Besides the inadequate or even non-existent quantitative and qualitative data on the various studies carried out by the transmission system operator (TSO) in this context, on the methodology of the cost-benefit analyses and on the hypotheses for the selected projects, this time period as provided for by the Royal Decree of 12 May 2024 is also largely inadequate to allow the CREG to validly exercise its powers.

After the federal development plan is approved by the minister, the CREG's powers are significantly limited. For example, the CREG has no binding prerogative to impose a revision of the plan. In fact, article 13, § 3, first paragraph, of the Electricity Act stipulates that the CREG can make recommendations to the minister, after consulting the TSO, to amend the federal development plan during its term, if the planned investments are insufficient. Based on these recommendations, the minister can oblige the TSO to amend the federal development plan. However, the CREG's opinions are not binding.

Therefore, the CREG cannot amend the federal development plan on its own initiative nor request the minister to amend it.

Once the federal development plan is approved, the CREG's tariff methodology must enable implementation of the plan by the system operator (Art. 12, § 5, 4° of the Electricity Act). In other words, the CREG's competence is limited to verifying whether or not the costs of the investments included in the development plan are reasonable, according to the criteria set out in its tariff methodology. Investment costs deemed reasonable are included in the tariffs of the transmission system operator.

⁵ The Royal Decree of 12 May 2024 laying down the procedure for the preparation, approval and publication of the hydrogen transmission system development plan and the electricity transmission system development plan replaces the Royal Decree of 20 December 2007 on the procedure for the preparation, approval and publication of the transmission system development plan, based on which the 2024-2034 Federal Development Plan was prepared and approved.

⁶ This was provided for in Article 6, second paragraph, of the above-mentioned Royal Decree of 20 December 2007 and was reprinted in Article 30, §1, of the replacement Royal Decree of 12 May 2024. The Royal Decree of 20 December 2007 even provided for a period of 30 days.

⁷ Most recently, in its [opinion \(A\)2445](#) of 15 September 2022 on the 2024-2034 Federal Development Plan.

⁸ Memorandum dated [9 July 2020](#) and [12 July 2024](#) for the attention of the 'formateur' of the federal government.

Based on its memorandum and the various findings explained in this report, the CREG puts forward several recommendations for changing its operating framework, in order to prevent this situation from recurring in the future and to ensure stable and effective regulation (chapter 9.2 Recommendations).

2. ANTECEDENTS

2. On 20 June 2024, the CREG informed the Minister for Energy, the Minister of the North Sea and DG Energy by letter regarding the budget increase to [CONFIDENTIAL: between € 6.5 and 7.5 billion] for MOG II.

3. On 10 September 2024, the CREG had consultations with Elia on the exponential budget increase of MOG II. The following topics were discussed at this meeting:

- communication of figures to the CREG: Elia should proactively communicate accurate figures to CREG in good time;
- mitigating measures or alternative options for the MOG II design;
- financing for an investment of [CONFIDENTIAL: between € 6.5 and 7.5 billion].

4. On 17 September 2024, two members of Parliament in the Energy, Environment and Climate Committee asked questions to the Minister for Energy regarding the MOG II's cost estimate. The minister replied as follows (freely translated):

*"The budget for the energy island is subject to increases related to inflation, but also to the evolution of material costs and the tight market of the supply chain for parts. This means that regardless of the selected design, we would have faced cost increases anyway, since both an island as well as platforms require the DC infrastructure."*⁹

5. On 19 September 2024, the Minister for Energy asked Elia and the CREG in a letter to jointly take all possible measures to control costs and minimise the impact on the transmission tariffs. She also referred to an integrated approach and recommended setting up a dedicated workgroup. The *Workgroup Welfare* and *Workgroup Tender* were set up to this end.

6. In the context of the Minister's questions and in preparation of the present report, the CREG organised several meetings with Elia on each topic:

- The Workgroup Welfare met on 8 and 21 October, 18 November, 13 and 19 December 2024. This workgroup discussed the elements covered in chapter 6;
- The Workgroup Tender met on 15 October, 6, 15 and 22 November and 2, 9 and 17 December 2024. This workgroup discussed the audit by DNV. The work and conclusions can be found in chapter 5 of the present report.

⁹ Free translation of : "Het budget van het energie-eiland is onderhevig aan stijgingen die te maken hebben met inflatie, maar ook met de evolutie van de materiaalkosten en met de krapte op de markt van de toeleveringsketen van onderdelen. Dat betekent dat we, ongeacht het gekozen design, aangezien zowel een eiland als platformen de DC-infrastructuur nodig hebben, te maken zouden hebben gekregen met kostenstijgingen".

7. On 22 October 2024, two members of Parliament asked questions to the Minister for Energy on "the cost estimate of the energy island" at the Energy, Environment and Climate Committee (freely translated):

"You admitted last time that costs have increased, but you did not say what the order of magnitude of the current estimates is. Meanwhile, the figure of €7 billion is increasingly being cited. Is it true that the cost estimates are approaching or exceeding 7 billion? Or is that incorrect?"¹⁰

The minister replied as follows (freely translated):

"It's difficult to say. I can't put a concrete figure on it. You also know the reason why, the fact that the construction of the island, the contract with TM EDISON, is ongoing. The system operator informs me that the costs are being monitored and are under control.

The cost for the AC part has already been made public. The total amount is €1.4 billion. This information is publicly available on Elia's website.

For the DC part, Elia is still negotiating with suppliers. Elia obviously has to respect all legal confidentiality obligations in this regard.

The award of the substation is planned for the end of 2024, and the cable by the end of Q1 2025.

The increase in costs is real, not only for the energy island but for the whole value chain. Indeed, the CREG has informed me that it is very concerned about this situation. I have also discussed this with Elia. Even though I don't actually have anything to do with it - it is between Elia and the CREG - I have sent a letter to the CREG with the clear request to limit the cost increase as much as possible via an integrated approach, by setting up a joint workgroup with Elia and the CREG, which looks at cost reduction and transparency using the drip-feed method. With regard to limiting the impact on transmission tariffs, they should examine together, proactively and in dialogue, what cost-saving measures can be taken. For example, staggered costs or other mechanisms that can ease the financial burden. [...]"¹¹

8. On 5 November 2024, there was an exchange of views on the increase in the cost of the Princess Elisabeth energy island with the Minister for Energy in the Energy, Environment and Climate Committee.

¹⁰ Free translation of : "U hebt vorige keer toegegeven dat de kosten gestegen zijn, maar u hebt niet gezegd wat de grootteorde van de huidige ramingen is. Ondertussen wordt alsmaar vaker het bedrag van 7 miljard euro genoemd. Klopt het dat de kostenramingen 7 miljard benaderen of overschrijden? Of klopt dat niet?"

¹¹ Free translation of : "Dat is moeilijk te zeggen. Ik kan daar geen concreet cijfer op plakken. U kent daarvoor ook de reden, namelijk de bouw van het eiland, het contract met TM EDISON, is lopende. De netbeheerder informeert me dat de kosten daarvan worden gemonitord en onder controle zijn.

De kostprijs voor het AC-gedeelte werd reeds publiek bekendgemaakt. Het gaat over een totaalbedrag van 1,4 miljard euro. Die informatie is publiekelijk beschikbaar op de website van Elia.

Voor het DC-gedeelte is Elia nog in onderhandeling met de leveranciers. Daar moet Elia uiteraard alle wettelijke confidentialiteitsverplichtingen respecteren.

De gunning van het substation staat gepland voor eind 2024 en de kabel eind Q1 van 2025.

De stijging van de kosten, niet alleen voor het energie-eiland maar voor de hele waardenketen, is reëel. De CREG heeft mij inderdaad laten weten dat zij zich daarover grote zorgen maakt. Ik heb daar ook met Elia over gesproken. Ook al heb ik daar eigenlijk niets mee te maken – dat is iets tussen Elia en de CREG –, heb ik een brief aan de CREG gericht met de duidelijke vraag om de kostenstijging zoveel mogelijk te beperken via een integrale aanpak, met de oprichting van een gezamenlijke werkgroep met Elia en de CREG, waarin met de kaasschaafmethode naar kostenreductie en transparantie moet worden gekeken. Met betrekking tot de beperking van de impact op de transmissietarieven moeten zij samen, in dialoog en proactief, onderzoeken welke kostenbesparende maatregelen ze kunnen nemen. Ik denk bijvoorbeeld aan gespreide kosten of andere mechanismen die de financiële lasten kunnen verlichten. [...]"

9. On 12 November 2024, there was a hearing in the Energy, Environment and Climate Committee on MOG II, with representatives from the boards of directors of Elia and the CREG. At this hearing, the CREG announced its report on the MOG II (with a detailed cost analysis) by the end of January 2025.
10. On 27 November 2024, a second hearing on the MOG II was held in the Energy, Environment and Climate Committee with the Minister for Energy and the representatives of the boards of directors of Elia and the CREG.
11. On 24 January 2025, the CREG's board of directors approved report (RA)2960 on the budget increases of the MOG II project, at first reading. Report (RA)2960 was then sent to Elia in order to identify any material errors and confidential elements.
12. On 30 January 2025, the CREG's board of directors approved the executive summary of report (RA)2960 and sent it to the system operator.
13. On 31 January 2025, the system operator responded by identifying the confidential elements in report (RA)2960 (including the executive summary) and adding its comments to the present report.
14. On 5 February 2025, the non-confidential version of the executive summary of report (RA)2960 (taking into account the confidential elements identified by Elia) was sent to the Minister for Energy and the Chairman of the Energy, Environment and Climate Committee.
15. On 18 February 2025, the CREG's board of directors approved the final version of report (RA)2960.

3. BACKGROUND TO THE MOG II PROJECT

3.1. GRID DESIGN MOG II

16. At the Ministerial Council of 20 April 2018, the federal government approved the Federal Energy Strategy. This legal framework should allow the federal government to achieve the 4 GW of offshore wind energy envisaged in the Interfederal Energy Pact by 2030 at the latest.

17. On 31 August 2018, the Ministerial Council approved the policy document "Tendering offshore wind farms as from 2020"¹². The government opted for a tender mechanism to achieve the following four objectives:

- renewable wind power capacity (min. 1.7 GW - max. 2.1 GW);
- maximum energy supplied to the grid (in GWh);
- minimal cost to consumers (directly or indirectly);
- energy grid support services (balancing, storage, etc.).

18. The Royal Decree of 22 May 2019 establishing the marine spatial plan for the period from 2020 to 2026 in the Belgian marine areas updates and completes the first marine spatial plan from 2014. The zone created in May 2004 for wind farms is not changed, and an additional second zone is created for offshore energy generation, the Princess Elisabeth Zone.

¹² Source: <https://news.belgium.be/nl/tendering-van-de-offshore-windmolenparken-vanaf-2020>

19. The Federal Development Plan 2020-2030, approved on 30 April 2019, included the following as regards the "Modular Offshore Grid - Phase 2": (freely translated)

“...

In the Belgian part of the North Sea, 1.7 to 2 GW of additional offshore wind generation capacity (as well as other forms of energy production) is currently being sought. Elia is already proving, with the first phase in the realisation of the Modular Offshore Grid, that bundling connections from offshore wind farms at sea is a more sustainable and efficient way to bring this generated energy ashore, than each developer building its own connection.

...

The offshore benchmark solution involves one or more offshore platforms, to connect and link the additional offshore energy from the different zones to the onshore transmission system with cable connections. This equates to a further extension of the modular offshore grid. The study also includes a section where the desirability of moving towards a hybrid project is considered. The benchmark solution for connection to the onshore transmission system is the new Stevin-Avelgem corridor ("Ventilus"), an essential condition for creating this receiving capacity.

...

The project is included in this development plan, on condition that the draft law amending the Electricity Act is approved, with a view to introducing a competitive bidding procedure for the construction and operation of generation facilities in marine areas, which outlines the main roles and responsibilities, including those of Elia a.o..

...”

20. The benchmark solution for the MOG II consisting of three platforms was fully prepared during 2019-2020. In the Taskforce Tendering Offshore¹³, the work related to the tendering of the Princess Elisabeth Zone and the connection of the Princess Elisabeth Zone through MOG II was closely followed up.

21. On 1 October 2020, the new federal government took office.

22. In October 2020¹⁴, Chris Peeters, then-CEO of Elia, stated in De Standaard that it would be much cheaper to simply build a connection towards the Belgian coast via platforms to connect the second wind zone (PEZ).

23. The detailed grid design file for the benchmark solution for the MOG II consisting of three platforms was ready to be submitted to the CREG for an opinion¹⁵ at the end of 2020. The grid design approval by the Ministers for Energy and the North Sea was initially scheduled for April 2021. This was in line with the planning set by the Taskforce Tendering Offshore.

24. At the Taskforce Tendering Offshore meeting on 28 January 2021, it was first reported that a new option was being looked at by Elia, an island. A specific taskforce meeting was organised on 12 February 2021 in which the energy island was presented. Prior to the meeting, the CREG submitted

¹³ The Task Force Tendering Offshore includes representatives from the Ministries for Energy and the North Sea, DG Energy, Department for the Marine Environment, MUMM, the CREG and Elia. At the meetings of this task force, the tendering of the Princess Elisabeth Zone is prepared by monitoring the legislative framework (follow-up of the necessary adaptations to the Electricity Act and drafting of the necessary royal decrees and ministerial orders), the environmental permit and the grid design of the MOG II.

¹⁴ Source: De Standaard, 21 October 2020, "Elia pioniert mee op zee".

¹⁵ Opinion based on Article 6/4, §2 of the law of 29 April 1999 on the organisation of the electricity market.

specific questions regarding the energy island and the comparison with the initial design consisting of three 700 MW platforms.

25. Choosing the island could only be justified if it did not cost more for electricity consumers than the option with the three 700 MW platforms that were needed to connect up to 2.1 GW in the Princess Elisabeth Zone¹⁶.

26. The energy island, called the Belgian Energy Hub, is located 40 km off the Belgian coast, just outside the Princess Elisabeth Zone. The intention was that the multifunctional island, with a total area of 10 ha, would combine transmission system activities with commercial activities. At the time, the usable area of 5 ha was divided as follows:

- 3 ha reserved for the Princess Elisabeth Zone connection ;
- 2 ha intended for commercial activities such as data centres or an "O&M one stop offshore shop".

27. During several workshops with Elia and the CREG (on 1 and 16 March 2021 and 6 and 21 May 2021), both the costs involved as well as the governance of the proposed concept were discussed. Based on the information provided by Elia during these meetings, the CREG could not conclude that the concept of an energy island would be cheaper than three platforms. Moreover, this comparison also had to take into account the sunk costs, namely Elia's expenses over the period 2019 to 2020 for personnel costs and consultancy for developing the benchmark solution consisting of three platforms.

28. In response to the outbreak of the Covid pandemic and subsequent economic recession, the EU adopted a Recovery Plan for Europe, under which specific projects that would generate economic growth for member states in the short and medium term could receive financial support in the form of a Recovery and Resilience Facility grant. This support was conditional on the projects contributing to the EU's energy and digital transition objectives. On 30 April 2021, Belgium submitted its "National Recovery and Resilience Plan" to the EC. The creation of a multifunctional island, which would bring together transmission facilities (3 ha) and other activities that met EU objectives (2 ha), was presented as a potentially successful project for Belgium in the context of the recovery plan.

29. At the Taskforce Tendering Offshore special meeting on 7 June 2021 regarding the status of the Belgian Energy Hub, the Taskforce was informed for the first time that informal enquiries had been made to Elia to connect a bigger capacity in the Princess Elisabeth Zone. It was confirmed that this seemed possible for a maximum of 3.5 GW combined with an interconnector.

¹⁶ Taskforce Tendering Offshore, special meeting of 12 February 2021, report, "2. Explanation of start conditions already identified":

"During the preparatory talks, several minimum conditions were already identified that must be met before the project can be launched. These are:

a. financial:

- i. the island must [have] a positive impact on energy tariffs compared to the three-platform scenario*
- ii. the grant from the RRF will not be used for the commercial activities part*
- iii. the commercial risks associated with the other activities will not be borne by the regulated part of ELIA*

b. construction engineering / environment:

- i. Efforts are made to ensure that as little sand as possible is used for construction (ambition: sand neutral)*

As regards the starting condition for the environment, we note that this is very limited. It is clarified that these are initial conditions that could already be identified, but that environmental efforts will not be limited to this condition. These preconditions are the big no-go principles that need to be identified."

30. On 16 June 2021, the Minister for Energy sent a letter to Elia with the request to examine the possibility of connecting more power (up to a maximum of 3.5 GW) than the 2.1 GW planned thus far in the Princess Elisabeth Zone. In addition, the minister asked Elia to propose a technical design to connect an energy hub in the form of an artificial island for up to 3.5 GW of offshore wind.

31. At the Taskforce Tendering Offshore meeting on 24 June 2021, the minister's letter to Elia was discussed. The appointment of a consultant for the LCOE study was discussed in order to determine the optimal capacity of the Princess Elisabeth Zone. The LCOE study would calculate the impact of increasing the density (and the associated increase in wake losses¹⁷) on the LCOE.

32. On 12 July 2021, a kick-off meeting was held with 3E¹⁸ and the members of the Taskforce Tendering Offshore, on 3E's study regarding the LCOE calculation. Prior to this kick-off meeting, the CREG transmitted its comments on 3E's tender and the proposed methodology to the Taskforce members by e-mail dated 7 July 2021.

33. On 14 July 2021, the CREG forwarded its comments regarding the proposed methodology of the LCOE study to the Taskforce Tendering Offshore. The CREG's main finding was that the study presented the following two effects, combined:

- the impact of capacity/density on the LCOE;
- the impact of connection mode (island vs. platforms, AC vs. DC) on the LCOE.

The actual aim of the study was to calculate the impact of density on the LCOE. Working with 69 scenarios made things look very complicated. A better approach would be to work in two stages:

- first, the optimal capacity in the Princess Elisabeth Zone should be calculated;
- based on this ideal capacity, a grid design could be proposed based on a trade-off between different criteria for two scenarios (x number of platforms vs one island).

34. On 18 August 2021, 3E's first results were discussed with the members of the Taskforce Tendering Offshore. At this meeting, the CREG expressed its concerns regarding 3E's (low) estimate of wake losses for the first time. CREG also referred to the study conducted by 3E in 2016 that examined the impact on the LCOE due to the different density of the Belgian offshore parks and Borssele. Indeed, this study estimated the wake losses and the impact on Full Load Hours to be much higher.

35. On 8 September 2021, members of the Taskforce Tendering Offshore received the draft version of 3E's report. The same day, the CREG submitted its comments to 3E, and the Ministries for Energy and the North Sea. At the Taskforce Tendering Offshore meeting on 9 September 2021, 3E explained its draft report.

36. On 15 September 2021, members of the Taskforce Tendering Offshore received 3E's final report. On 16 September 2021, the CREG submitted its comments to the Ministries for Energy and the North Sea. The CREG also discussed its comments at the Taskforce Tendering Offshore meeting on 23 September 2021.

¹⁷ Wake losses are losses that reduce electricity production, whereby a distinction is made between internal and external wake effects. Internal wake effects refer to the interaction between wind turbines within the same offshore farm. A wind turbine extracts energy from the wind and uses it to produce electricity. This means that the energy in the wind decreases and the turbines behind the first turbine in the row consequently produce less energy. The higher the density of wind turbines within an offshore farm, the greater these internal wake effects will be and the lower the electricity production. External wake effects refer to the interaction between wind farms. If wind farms are close to each other, the wind cannot recover between them. In other words, the wake effect will then be greater and the electricity output will therefore be lower.

¹⁸ This consultant was appointed by the DG Energy at the request of the Ministry for Energy.

37. In its letter dated 11 October 2021 to the Minister for Energy regarding the 3E study "*LCOE offshore wind in the Princess Elisabeth Zone*", the CREG questioned the impact of the planned increase in offshore capacity and the impact on the grid design.

38. On 15 October 2021¹⁹, the Ministerial Council took note of the planned activities and objectives in the development of the Princess Elisabeth Zone and the construction of the energy hub. The Ministerial Council validated the objective of a minimum production of 3.15 GW and a maximum production of 3.5 GW in the Princess Elisabeth Zone as well as the concept of connecting wind energy via an energy island.

39. On 18 October 2021, the CREG received the "design for the extension of the Modular Offshore Grid" from Elia Transmission Belgium, for its opinion. Elia examined and compared three variants for the connection of offshore wind generation (3.15 - 3.5 GW) in the Princess Elisabeth Zone. These three variants are:

- variant 1: a connection through five 700-MW AC platforms;
- variant 2: a connection through AC and HVDC on an artificial island;
- variant 3: a connection through three AC platforms of 700 MW and one HVDC platform of 1,400 MW.

Elia compared the three variants based on the following criteria:

- investment cost;
- maintenance cost;
- cable lengths;
- timing;
- availability and reliability;
- flexibility with respect to auction strategy and number of parcels for wind farms;
- environmental impact and permit procedures;
- technical complexity;
- future prospects.

40. In Opinion (A)2302²⁰ on Modular Offshore Grid phase 2: "design for the extension of offshore transmission grid" of 26 November 2021, the CREG made its comments on the grid design. As such, it noted that in order to make a full comparison between the three variants for the proposed offshore grid concept, the evaluation of certain criteria of the analysis of the variants should have been more substantiated with quantitative data. A qualitative evaluation of the criteria was not sufficient and above all meant that the CREG could not verify Elia's analysis or the conclusions of these analyses. In particular, the CREG would have preferred that the assessment of the criteria on maintenance costs, availability and reliability and future prospects was substantiated with quantitative data.

¹⁹ Source: Ministerial Council of 15 October 2021, "Energie: productie-capaciteit van de Prinses Elisabeth-zone in de Noordzee",

[Energie: productiecapaciteit van de Prinses Elisabeth-zone in de Noordzee | News.belgium](#)

²⁰ [Opinion \(A\)2302](#) on the "Modular Offshore Grid phase 2: design of the extension of the offshore transmission grid"

Moreover, the CREG noted that the investment cost of the three variants was estimated by Elia as follows.

Table 1 – CAPEX overview per variant

CAPEX overview per variant					
Variant 1: 5 * 700 MW platforms	M EUR	Variant 2: AC/HVDC island	M EUR	Variant 3: AC/HVDC platforms	M EUR
Offshore AC-platforms	600	Artificial island (environmental measures incl.)	450	Offshore AC platforms	360
Offshore AC high and low voltage equipment	115	AC substation offshore on the island	100	Offshore AC high and low voltage equipment	70
AC export and interconnexion cables	990	Offshore AC high and low voltage equipment	70	Offshore HVDC platform	720
Onshore AC cables	140	HVDC conversion station offshore island	335	AC export cables	600
Onshore AC high and low voltage cables	35	AC export cables	600	DC export cables	115
Measures for AC system stability	100	DC export cables	100	AC onshore cables	50
Other (PM, insurance, risk, overhead, ...)	240	AC onshore cables	50	DC onshore cables	25
		DC onshore cables	25	Onshore HVDC conversion station	210
		Onshore HVDC conversion station	210	Onshore AC high and low voltage equipment	30
		Onshore AC high and low voltage equipment	30	Other (PM, insurance, risk, overhead, ...)	260
		Other (PM, insurance, risk, overhead, ...)	235		
CAPEX	2220	CAPEX	2205	CAPEX	2440

Elia did not include the cost of the 66 kV AC inter array cables between the wind turbines and the connection to the transmission grid (platform or island) in the CAPEX comparison. While these costs are indeed not borne by Elia (and therefore not passed on in the transmission tariffs), they should definitely be included in the cost comparison. After all, these costs will be included in the tendering by the offshore wind operators and therefore paid by society (either through electricity consumers or public funds). As Elia itself pointed out, the number of km of inter array cables strongly depends on the scenario, which means that each scenario has a different social cost. The CREG already passed this observation²¹ on to the members of the Taskforce Tendering Offshore on 23 February 2021 after receiving the report of the special meeting of 12 February 2021.

If the cost of the inter array cables is included, the sum of the two components (based on the information in the file) leads to the following result.

²¹ Email dated 23/02/2021 from a CREG employee (freely translated):

"If the choice of an island is justified based on "a positive impact on energy tariffs compared to the three-platform scenario" (as mentioned in point 2a of the report), the following principles should be taken into account:

- Energy tariffs = transmission system tariffs (where Elia's costs are passed on to consumers) + offshore surcharge (the grant - which is based on costs - is passed on from the offshore farms to consumers)
- All costs impacted by choosing the island versus 3 platforms should be included, and therefore also those for the farms (e.g. infield cables). These costs should be fully reflected and therefore not just be additional or less costs.
- The costs or income from commercial activities are not included as they must not have any impact on transmission system tariffs.

Specifically, this means that the following comparison should be made.

miljoen EUR	Eiland (1)	3 platformen (2)	Delta = (2)-(1)
Constructie Eiland* of platform			
RRF subsidie	-100	0	
HV & LV equipment			
Kabels			
Survey & UXO			
Project Staff + management			
Construction insurances			
CAPEX Elia	-100	0	100
Infieldkabels offshore parken			0
Totale kostprijs	-100	0	100
Sunk costs "3 platformen"			
Totale besparing voor de consument door keuze voor eiland			100

*= kosten constructie eiland ten laste van de gereuleerde activiteit

Table 2 - Overview of CAPEX per variant incl. cost of inter array cables

	Variant 1: 5 * 700 MW platforms	Variant 2: AC/HVDC island	Variant 3: AC/HVDC platforms
CAPEX (M EUR)	2220	2205	2440
# km inter-array cable	438	643	440
cost inter-array cable (M EUR/km)	0,65	0,65	0,65
cost inter-array cable (M EUR)	285	420	285
Total CAPEX inter-array cable incl. (M EUR)	2505	2625	2725

Based on the quantitative calculations in the file, the CREG could not endorse Elia's claim that the most cost-efficient connection is a connection via an island. However, thanks to the EU Recovery and Resilience Facility grant of up to €100 million (hereinafter "RFF grant"), the additional costs could potentially be limited.

The CREG also noted that the island variant was at the same level as AC platforms in terms of technical complexity, despite the fact that Elia also stated that "*the development and construction of an artificial island (first of a kind in the North Sea) is more complex than an AC platform*". As such, the CREG pointed out in its opinion (in § 24) (freely translated):

"...

The CREG underlines that there is a high cost impact if the island is not built in time. If the island is not completed before 31 August 2026, Belgium will lose the Recovery and Resilience Facility grant of up to €100 million. Therefore, the CREG believes that the financial impact of not being awarded an RFF grant for the island due to late completion should not be passed on to the grid user.

..."

41. On 23 December 2021, the ministerial council approved a draft ministerial decree establishing the design for the extension of the Modular Offshore Grid for the Princess Elisabeth Zone.

42. On 26 January 2022, the CREG received the "File for the evaluation by the CREG of electricity infrastructure investments and associated higher risks" from Elia. In this file, Elia identified a series of risks for the MOG II grid design. As these risks were not identified nor quantified in the file "design for the extension of the Modular Offshore Grid" dated 18 October 2021, the CREG informed the Minister for Energy and the Minister for the North Sea of these new developments by letter dated 4 February 2022. Elia identified additional risks for the MOG II, amounting to [CONFIDENTIAL] above the already budgeted amount of € 2.2 billion²². For the amount of € [CONFIDENTIAL], € [CONFIDENTIAL] risks were linked to the HVDC technology and € [CONFIDENTIAL] to the island itself. In this letter the following was also mentioned:

"To the extent that the current grid design chosen for MOG II (an energy island where a 2.1 GW connection is envisaged through three AC substations and a 1.4 GW connection through an HVDC substation) will be confirmed in the next Federal Development Plan, the CREG has to accept the costs (and any additional costs due to higher risks) for this investment. The CREG can only verify the (investment) costs as to their reasonableness. Consequently, the uncertainty regarding the various cost items and the associated financial risk are borne entirely by the grid user - and not by Elia."

43. At the Taskforce Tendering Offshore meeting of 11 February 2022, the CREG raised this problem and explained its letter of 4 February 2022.

²² The budget of € 2.2 billion is included in the file "design for the extension of the Modular Offshore Grid" dated 18 October 2021 and includes a risk contingency of 12 %.

44. In Opinion (A)2445 on the draft 2024-2034 development plan of Elia Transmission Belgium of 15 September 2022, the CREG stated the following (freely translated):

"In addition, the CREG would like to mention that the CBAs are still a black box for the CREG and it is difficult to verify the figures. ... The CREG recalls that it has to accept the costs for investments approved by the Federal Development Plan. The CREG can only verify the (investment) costs as to their reasonableness"²³.

45. The 2024-2034 Federal Development Plan was approved by the ministerial decree of 5 May 2023. The MOG II grid design was confirmed in this federal development plan, and consisted of an energy island with a 2.1 GW connection through three AC substations and a 1.4 GW connection through an HVDC substation.

46. On 7 September 2023, the ministerial decree approving the design for the extension of the Modular Offshore Grid, in accordance with article 6/4 of the law of 29 April 1999 on the organisation of the electricity market, was approved. As proposed by Elia, an energy island would be the most optimal way to achieve the connection of up to 3.5 GW of offshore wind power from the Princess Elisabeth Zone, with a 2.1 GW connection through three AC substations and a 1.4 GW connection through an HVDC substation. The Ministerial Council already took a first decision on this grid design on 15 October 2021 and 23 December 2021, respectively.

47. In Opinion (A)2652 of 5 October 2023, the CREG expressed its regret that there was no information on decommissioning costs in the file "Modular Offshore Grid phase 2: design of the extension of the offshore transmission grid" and that it was therefore not included in the analysis comparing the different variants (island versus platforms).

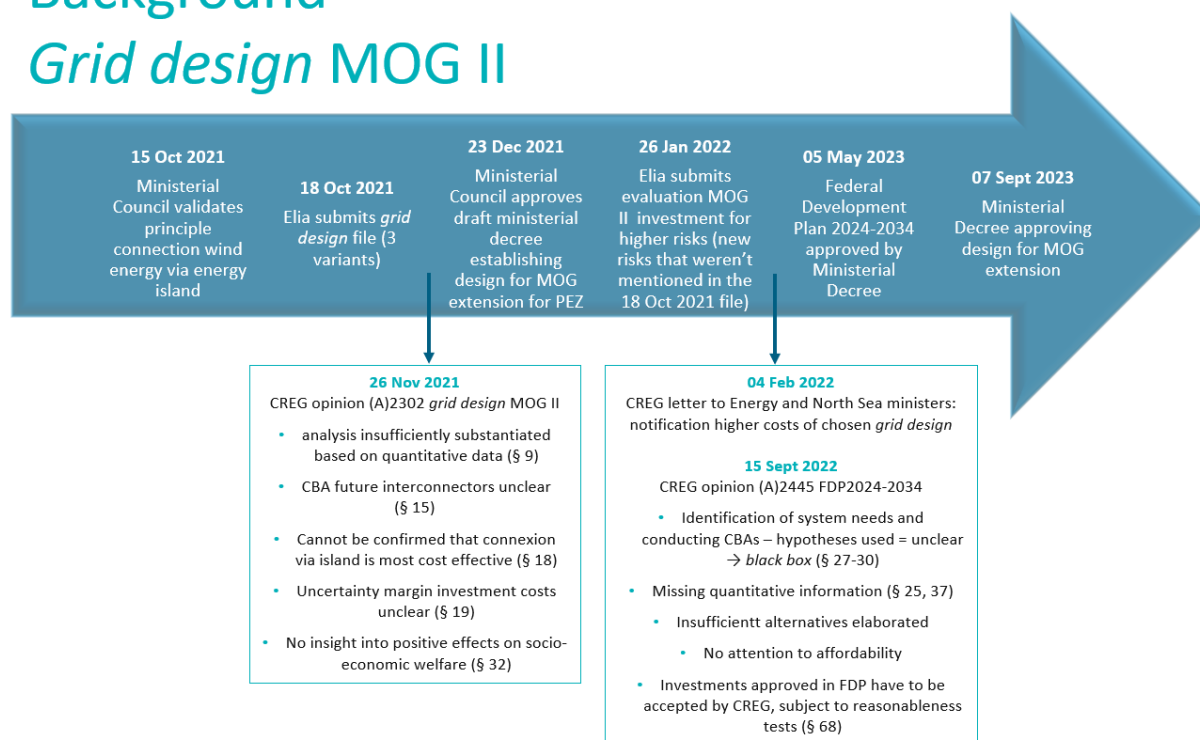
48. The following figure summarises the key elements.

²³ Free translation of: "Voor zover het nu gekozen grid design van MOG II (een energie-eiland waarbij een aansluiting van 2,1 GW wordt voorzien via drie AC-onderstations en een aansluiting van 1,4 GW via een HVDC-station) zal worden bevestigd in het volgende Federale Ontwikkelingsplan, dan moet de CREG de kosten (en eventuele meerkosten door hogere risico's) voor deze investering aanvaarden. De CREG kan enkel de (investerings)kosten controleren op redelijkheid. De onzekerheid op het niveau van de verschillende kostenposten en het bijhorende financiële risico worden bijgevolg volledig door de netgebruiker - en niet door Elia - gedragen."

Figure 1 - Grid design MOG II – background

Background

Grid design MOG II



49. Regarding the Recovery and Resilience Facility grant, the CREG pointed out above (in § 35) that (freely translated):

- the additional cost of an island (compared to platforms) was potentially limited as the €100 million RFF grant had been secured;
- according to Elia, the development and construction of an artificial island would be a first of a kind in the North Sea; however, the island variant was rated the same as the AC platforms in terms of technical complexity. Therefore, the CREG stated in its Opinion (A)2302 that there would be a significant cost impact should the island not be built in time. If the island was not built by 31 August 2026, Belgium would lose the RFF grant of up to €100 million. The CREG considered that the financial impact of not obtaining an RFF grant for the island on account of late completion should not be passed on to the grid user.

On 10 January 2025, the CREG received a copy of Elia's letter to DG Energy, in which Elia reported the following:

*"it seems useful to point out that the contractor has informed us, based on informal discussions, that there is a risk of a delay of several months in the completion of the project."*²⁴

²⁴ Free translation of "Cependant, il nous semble utile de vous préciser que, sur base d'échanges informels avec le contractant, ce dernier nous a fait part qu'il existe un risque de retard de plusieurs mois pour la réalisation du projet."

This letter formalises the rumours informally circulating in the sector that the island might not be completed by 31 August 2026 and consequently the milestone for obtaining the RFF grant would not be met.

3.2. EVOLUTION OF THE GRID DESIGN AFTER 2021: CHANGE OF SCOPE

50. Elia stated that it had made three major changes between the presentation of the grid design file (late 2021) and the FID decision (2023)²⁵:

- the size of the island was increased from 5 to 6 ha;
- an electrical link will be made between the AC and DC substations using additional 220 kV AC infrastructure, with the aim of enabling the system to operate in a single-node configuration;
- the capacity of the HVDC link (converter and cables) was increased from 1.4 to 2 GW to bring it in line with market standards for 525 kV bipolar HVDC links.

51. Chapter 4 of this report assesses the impact of these changes on costs and their relevance.

3.3. INCREASE IN THE ESTIMATED COST OF THE GRID DESIGN

52. When the grid design was approved at first reading on 23 December 2021, the cost of this investment was estimated at **€2.2 billion**²⁶.

53. In the context of Elia's 2024-2027 tariff proposal (in the first quarter of 2023), the CREG was confronted with a significant budget increase for the MOG II for the first time, i.e. from €2.2 billion to €3.6 billion. On 28 April 2023, the CREG explained this to the Ministry for Energy during a presentation on the main evolutions of the 2024-2027 tariff proposal.

54. On 5 March 2024, Elia's board of directors was informed of the latest budget estimate of **[CONFIDENTIAL]**. The board of directors then gave the executive management the following orders²⁷:

"[CONFIDENTIAL]".

55. Elia's executive management did not contact the CREG in March 2024 (see § 66). The CREG received Elia's 2023 tariff report on 28 February 2024. As part of the additional information, the CREG requested an update of the CAPEX (budget estimate) for the MOG II on 22 March 2024.

56. On 12 April 2024, the CREG received a response from Elia showing that the budget had increased to **[CONFIDENTIAL: but an amount considerably lower than the amount communicated to Elia's Board of Directors on 5 March 2024, cf. § 54]**.

57. On 25 April 2024, the CREG asked additional questions to Elia regarding this budget increase.

²⁵ Final Investment Decision, January 2023. The final investment decision refers to the stage in the project planning process when a decision is made to make a major financial commitment. This FID point is reached when a large order is placed for equipment or when EPCI contracts are signed. Elia awards EPCI contract for world's first energy island to DEME and Jan De Nul - press release 28/02/2023 https://www.elia.be/en/press/2023/02/20230228_epci-contract-energy-island-to-deme-and-jan-de-nul

²⁶ The cost of the different variants was included in the dossier submitted by Elia on 18 October 2021.

See § 35.

²⁷ Source: Minutes of the Board of Directors of 5 March 2024, chapter 5 Approvals, (iii) for package 4; substations with HVDC technology

58. On 13 May 2024, the CREG discussed the questions from 25 April 2024 during a meeting with Elia. For the first time, the amount of [CONFIDENTIAL: between € 6.5 and 7.5 billion] was then mentioned to the CREG.

59. On 31 May 2024, the CREG's board of directors sent an official letter²⁸ to Elia requesting further clarification regarding the figure of € [CONFIDENTIAL: between € 6.5 and 7.5 billion]. Elia replied on 11 June 2024.

60. On 20 June 2024, the CREG asked Elia by letter to explore alternatives. The same day, the Minister for Energy, the Minister for the North Sea and DG Energy were also notified by the CREG in a letter about the budget increase to [CONFIDENTIAL: between € 6.5 and 7.5 billion]²⁹.

61. On 11 July 2024, Elia sent a letter explaining why it believed the alternatives were not viable.

62. By letter dated 19 July 2024 to the CREG (received 23 July 2024), the Minister for Energy expressed her concerns. In this letter, the minister asked the CREG to quantify the MOG II benefits.

63. On 19 August 2024, the CREG replied as follows, among other things (freely translated):

"In your letter, you state that the CREG only emphasises the cost aspect without considering the societal interest or the broader context of the energy transition, energy security of supply and the affordability of the energy bill. Given the comments and reservations expressed by the CREG prior to the government's decision on the MOG II design, the CREG does not understand how you come to this conclusion. Without going into detail on the various interventions made by the CREG, the CREG would like to refer in particular to its Opinion (A)2302 of 26 November 2021 on the design for the Modular Offshore Grid phase 2. This opinion clearly highlighted the glaring lack of information and quantitative data to be able to provide a motivated opinion. In that same opinion, the CREG also stated that the potential benefits could not be evaluated (quantitatively) on the basis of the submitted file. On 7 September 2023, the MOG II design was approved by ministerial decree. This ministerial decree makes no mention of additional socio-economic analyses carried out after the CREG's opinion. Therefore, the CREG does not have the quantitative cost-benefit analysis that supported the government's decision. Whatever the result of this cost-benefit analysis may have been, Elia's estimate that the initial budget might triple has a significant impact on it. By its letter, the CREG wanted to inform you of this cost evolution. It is impossible for the CREG to re-evaluate the benefits in a relevant manner without disposing of the initial cost-benefit analysis."³⁰

²⁸ Reference of the letter: 2024/P154/V280-CDC05.31

²⁹ Reference of the letter to the Minister for Energy: 2024/P166/V311-CDC06.20

Reference of the letter to the Minister for the North Sea: 2024/P167/V312-CDC06.20

Reference of the letter to DG Energy: 2024/P168/V313-CDC06.20

³⁰ Free translation of "In uw brief stelt u dat de CREG enkel het kostenaspect benadrukt zonder oog te hebben voor het maatschappelijk belang of de bredere context van de energietransitie, de energiebevoorradingszekerheid en de betaalbaarheid van de facturen. De CREG begrijpt niet hoe u tot deze conclusie komt, gezien de opmerkingen en bedenkingen die de CREG geuit heeft voorafgaand aan de beslissing van de regering over het MOGII-design. Zonder in detail te treden over de diverse interventies van de CREG wenst de CREG vooral te verwijzen naar haar advies (A)2302 van 26 november 2021 over het ontwerp van het Modular Offshore Grid fase 2. Dit advies benadrukte duidelijk het flagrante gebrek aan informatie en aan kwantitatieve gegevens om een gemotiveerd advies te kunnen verstrekken. De CREG stelde in hetzelfde advies ook dat de mogelijke voordelen niet (kwantitatief) konden geëvalueerd worden op basis van het ingediende dossier. Op 7 september 2023 werd per ministerieel besluit het ontwerp voor MOGII goedgekeurd. Dit ministerieel besluit maakt geen melding van bijkomende socio-economische analyses die werden uitgevoerd na het advies van de CREG. De CREG beschikt derhalve niet over de kwantitatieve kostenbatenanalyse die de beslissing van de regering ondersteunde. Wat het resultaat van deze kostenbatenanalyse ook moge geweest zijn, de inschatting van Elia dat het initiële budget mogelijks zal verdrievoudigen, heeft hierop een aanzienlijke impact. De CREG heeft door middel van haar brief u in kennis willen stellen van deze kostenevolutie. Het is onmogelijk voor de CREG om op een relevante manier de baten te herevalueren, zonder te beschikken over de initiële kostenbatenanalyse."

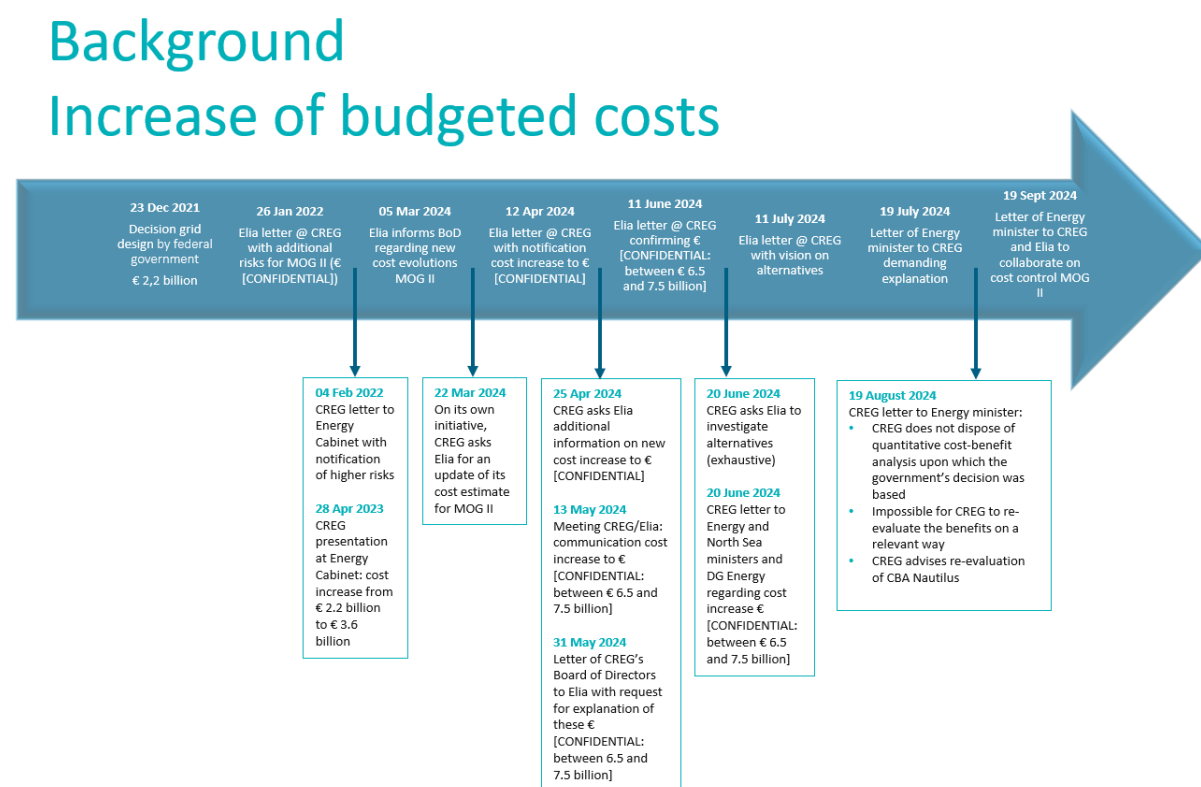
64. On 19 September 2024 the Minister for Energy asked Elia and the CREG, by letter received on 20 September 2024, to jointly take all possible measures to control the costs and to minimise the impact on the transmission system tariffs. The minister also referred to an integrated approach consisting of the following four elements:

- a detailed analysis of costs ("drip-feed method"); an external independent audit should probably be carried out in order to allow for an objective evaluation of the cost structure;
- the impact on transmission tariffs should be limited;
- a strengthening of European industrial policy;
- a comprehensive benefit analysis;

and a recommendation to set up a dedicated work group.

65. The following figure summarises the above-mentioned elements.

Figure 2 - Background to the increase in the estimated costs [NON-CONFIDENTIAL VERSION]



66. The CREG only took note of the question from Elia's board of directors (see § 49) after receiving the minutes of the board of directors dated 5 March 2024, on 27 October 2024.

67. It was only upon its own request (see paragraphs 50 - 53) that the CREG was informed of the very significant budget increase in May 2024.

During the discussion with Elia on 13 December 2024, the best estimate for the CAPEX for the MOG II was increased to **[CONFIDENTIAL: between € 7 and 8 billion]**³¹. Based on this figure, Elia calculated the Net Present Value of the various alternatives, which is discussed in chapter 6.

4. ESTIMATE OF THE IMPACT OF THE MOG II ON ELECTRICITY BILLS

68. European legislation³² establishes the principle of cost reflectivity, according to which all reasonable costs related to the operation of the transmission system must be passed on to grid users through transmission tariffs.

Article 12, § 2, 1st paragraph of the Electricity Act stipulates that following structured, documented and transparent consultation with the system operator, the CREG must establish the tariff methodology to be used by this system operator for drawing up its tariff proposal.

Article 12, § 5 of the Electricity Act further describes the guidelines that the CREG must comply with when establishing the tariff methodology. For the present discussion, the following guidelines are relevant in particular:

2° the tariff methodology must make it possible to efficiently cover all the costs that are necessary or efficient for the performance of the legal or regulatory obligations incumbent on the system operator, as well as for performing its activity of operating the transmission system or systems with a transmission function.

4° the tariff methodology allows the balanced development of the transmission network and networks with a transmission function, in accordance with the system operator's development plan referred to in Article 13 and the investment plans as approved, where applicable, by the competent authorities;

9° the normal remuneration of capital invested in the regulated assets should allow the system operator to make the necessary investments to perform its tasks;

22° with regard to the extensions of installations or new installations for the transmission of electricity, recognised as being of national or European interest, the tariff methodology referred to in § 2 may provide for specific provisions relating to the remuneration of the capital necessary to finance them which is more favourable than the normal remuneration of the capital referred to in § 5, 9°, as well as specific provisions relating to covering costs in order to promote their realisation and to enable their long-term development. Investments recognised as being of national or European interest are those realised by the system operator which contribute to national security of supply and/or optimal operation of cross-border interconnectors with, if applicable, phase shifting transformers, and therefore which facilitate the development of the national and European internal market or which contribute to the national reception of

³¹ According to the latest available data, the investment costs of the MOG II project have increased from € 2.2 billion to almost € [CONFIDENTIAL: between € 7 and 8 billion], which amounts to a multiplier effect of 3.5.

³² Article 18(1) of Regulation (EU) 2019/943.

production based on renewable energy sources which are either directly connected to the transmission network or indirectly through the distribution networks. Investments of national or European interest concern the installations which:

- *strengthen existing connections or create new connections of the grid operated by the grid operator using direct current (DC) technology;*
- *reinforce existing connections or create new connections operated by the system operator located in marine areas over which Belgium exercises its jurisdiction;*
- *strengthen existing cross-border interconnectors or create new cross-border interconnectors or that result from expanding the capacity of these interconnectors;*

23° (...) the financial burdens (...) are not subjected to an enabling regulation."

The transmission tariffs, as defined in the tariff methodology established by the CREG³³, therefore allow to cover, in particular, depreciation and financial costs, including the remuneration of the system operator, which includes a fair margin for the remuneration of the capital invested in the grid.

This fair margin is calculated each year by multiplying a percentage of the return by the value of the regulated asset base (RAB). This RAB is increased each year depending on the investment expenditure in the network, in particular that related to the MOG II.

Therefore, the increase in investment expenditure for the MOG II has a direct impact on the evolution of financing and depreciation costs related to this infrastructure and, by extension, on the level of transmission tariffs that will be charged to end consumers over the next six decades. Indeed, the tariff methodology stipulates that the artificial island for the MOG II project should be depreciated over 60 years.

Assuming that interest rates in the financial markets remain stable, financing and depreciation costs will gradually increase in line with the investments made, until the full infrastructure is commissioned. Once commissioned, these costs will gradually decrease over the next six decades according to the recorded depreciation. Since the electricity infrastructure of the MOG II will be depreciated over a maximum period of 33 years and only the artificial island will be depreciated over a period of 60 years, these costs will fall faster in the first three decades than in the subsequent three decades.

Under the assumption that the investment in the MOG II remains limited to € 7 billion - which is a low estimate if the DC part is retained - and that the full infrastructure is commissioned in 2034, the financing and depreciation costs related to the MOG II to be covered by the transmission tariffs would amount to around €629.5 million in 2034. Calculations made separately by both Elia and the CREG show that, without changes to the rules for allocating these costs, they would correspond to:

- €7.7/MWh (about €27/year) for a residential consumer, i.e. an increase in transmission tariffs of approximately 67 % compared to the transmission tariffs applicable in 2024;

³³ CREG, Decision (Z)1109/11 establishing the tariff methodology for the electricity transmission system and electricity grids having a transmission function for the regulatory period 2024-2027, 30 June 2022.

- €3.5/MWh for a large industrial consumer directly connected to the Elia 380/220/150 kV transmission grid, i.e. an increase of approximately 78 % in the transmission tariffs compared to those charged to these customers in 2024.

If the investment in the MOG II *in fine* turns out to be more than € 7 billion, the above amounts would have to be increased on the basis of a rule of three. For example, if the investment in the MOG II *in fine* were to be € 8 billion, the above amounts relating to the impact on the end consumer would have to be increased by 14.3 %³⁴.

While the above-mentioned financing and depreciation costs represent the bulk of the costs associated with the MOG II, in order to estimate the total impact of the MOG II on the electricity bills of end users, two categories of costs that are not directly affected by the increase in investment costs also need to be taken into account, namely the decommissioning provision and the operating costs:

- the decommissioning provision was set at € 381 million in constant value on 1 June 2023 in the Royal Decree of 21 March 2024 granting a domain concession to Elia Asset NV for the construction and operation of electricity transmission installations in the Princess Elisabeth Zone. This royal decree stipulates that this provision will be made in the form of an individualised provision in the balance sheet as from the first year of commissioning of the electricity transmission installations. This can therefore be estimated at € 25 million/year;
- the level of operating costs is more difficult to estimate as it will depend on contracts still to be concluded, and the amount will fluctuate significantly from year to year depending on the maintenance work to be carried out. Elia currently estimates these operating costs at € 74 to 97 million/year.

Assuming that the sum of the decommissioning provision and operating costs is on average € 110.5 million/year³⁵, the above-mentioned amounts relating to financing and depreciation costs would have to be increased by a further 17.6 %³⁶ to arrive at the total impact of the MOG II on the bill of the end consumer. This corresponds to an additional € 1.35/MWh (€ 4.7/year) for a residential consumer and an additional € 0.6/MWh for a large industrial consumer directly connected to the Elia 380/220/150 kV transmission grid.

5. DETAILED ANALYSIS OF THE MOG II INVESTMENT

69. As appears from the historic background, the CREG expressed critical comments both on the justification of the choices made regarding the technical design and on the financial consequences of these choices, at various times and during various steps in the investment process. The CREG did this proactively, constructively and within its powers.

³⁴ = 8 / 7

³⁵ = (2* 25 + 74 + 97) / 2

³⁶ = 110.5 / 629.5

70. During a contact with the Danish energy regulator DUR³⁷, Danish professor³⁸ Bent Flyvbjerg's book "*How Big Things Get Done*" was brought up.

In *How Big Things Get Done*, the authors³⁹ examine mistakes that cause (big) projects to fail and they identify a number of research-based principles that make projects successful. Drawing on a wide range of examples, the book identifies potential pitfalls in developing projects and also provides a number of answers, mostly summarised in heuristics, to identify and avoid these pitfalls. This chapter refers to *How Big Things Get Done* in a number of places to indicate which pitfalls in the MOG II project may have been insufficiently thought through and where the possible solutions lie.

5.1. MOG II DESCRIPTION

71. A detailed description of the MOG II project can be found in section 4.2.1 of the 2024-2034 Federal Development Plan.⁴⁰

The MOG II has two objectives. First, it aims to integrate wind energy production from the Princess Elisabeth Zone into the Belgian and European electricity grid. Developing this zone will allow to install between 3.15 and 3.5 GW of wind capacity. Construction of the wind farms is planned in three parcels, with commissioning initially scheduled between 2028 and 2030⁴¹. The wind turbines will be directly connected to the energy island via medium-voltage cables (66 kV). Elia will then be responsible for converting this to high-voltage (220 kV) and exporting the electricity to the coast.

The second objective of the MOG II is access to renewable energy produced elsewhere in the North Sea, thanks to interconnectors. The interconnectors provided for in the development plan are the Nautilus project, which connect Belgium to the UK, and the TritonLink project, which connects Belgium to Denmark.

To ensure these functions, the approved MOG II design consists of the construction of an artificial island that will house the offshore electricity transmission infrastructure. The island will be connected to the coast by high-voltage alternating current (HVAC) and direct current (HVDC) cables. The diagram in Figure 3 illustrates this concept.

³⁷ Danish Utility Regulator

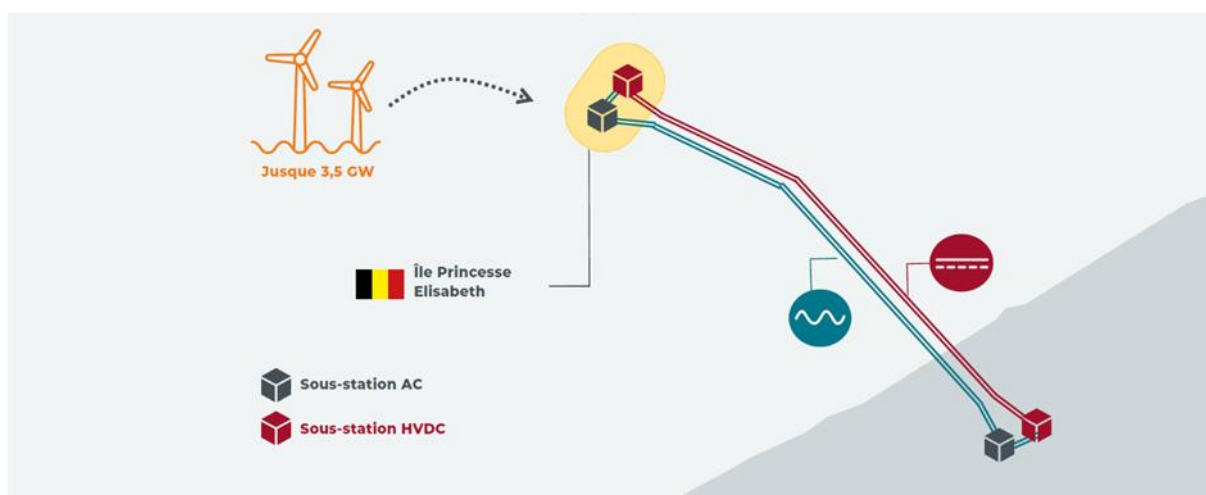
³⁸ Professor in economic geography associated with: University of Oxford, Delft University of Technology and the IT University of Copenhagen.

³⁹ *How Big Things Get Done*, Bent Flyvbjerg and Dan Gardner, Macmillan, UK 2023

⁴⁰ Source: <https://www.elia.be/nl/infrastructuur-en-projecten/investeringsplannen/federaal-ontwikkelingsplan-2024-2034>

⁴¹ Commissioning of the HVDC system is not planned until the summer of 2032 according to the latest Elia planning which the CREG has at its disposal.

Figure 3 - Schematic representation of the MOG II ⁴²



72. According to the grid design document dated 15 October 2021, confirmed by ministerial decree, the total transmission capacity of the MOG II, which is 3.5 GW, is split into 2.1 GW linked to alternating current (AC) infrastructure and 1.4 GW linked to direct current (DC) infrastructure. The main reason for using DC technology is the need to accommodate interconnectors: these offshore interconnectors can only use DC technology on account of the long distances to be covered.

73. Elia divided the MOG II into five packages, for which contracts were concluded separately.

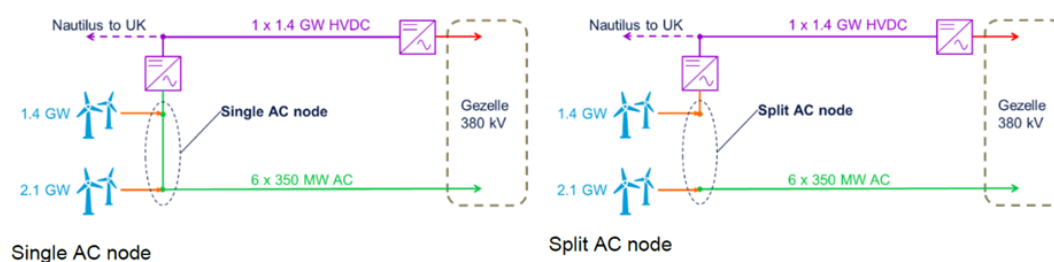
- 1) **The artificial island** includes the civil engineering works for the construction of a support based on caissons for the offshore transmission infrastructure;
- 2) **The AC substations** include AC transmission infrastructure. The main elements are 220 kV substations, 66 kV cabins, 220 kV / 66 kV transformers and shunt reactors for the reactive power. The offshore electrical infrastructure will be installed in modules, with the exception of transformers and shunts that will be installed directly on the island. A module is a steel structure built onshore and then transported and installed on the island.
- 3) **The AC cables** consist of six 220-kV AC cables that connect the island to the coast, and run approximately 55 km into the sea.
- 4) **The HVDC substations** include an offshore AC/DC converter and an onshore AC/DC converter, with a voltage of 525 kV. As is the case for the AC infrastructure, the offshore converter will be built in a module;
- 5) **The 525-kV DC cable system** that connects the offshore AC/DC converter to the onshore AC/DC converter.

74. However, these five packages are not completely separate from each other. Indeed, an electrical connection is planned between the offshore AC/DC converter and the AC stations. This connection theoretically allows the system to operate in a single node configuration. If this connection is not used, it will be a split node operation, as shown in Figure 4.

⁴² Source: Federal Development Plan (p. 187):

<https://www.elia.be/en/infrastructure-and-projects/investment-plan/federal-development-plan-2024-2034>

Figure 4 - Configuration of single node (left) and split (right) node⁴³



75. Single node means that the offshore wind turbines, the offshore HVDC converter and the AC export cable are all electrically connected. This provides more operational flexibility in the system, for example in terms of the use of the interconnector. On the other hand, additional investment costs have to be incurred and more transformers are needed.

76. Elia has concluded contracts for the electricity infrastructure that will enable the connection between the AC and DC parts and thus the single node operation. This infrastructure will therefore be installed on the island. Initially, however, the system will be operated in split node mode, mainly due to technical and market integration issues that have not yet been resolved⁴⁴. The layout of the energy island and the implantation of the infrastructure are shown in Figure 5.

Figure 5 - Layout of Princess Elisabeth island⁴⁵



⁴³ Source: DNV Report, Technical and economic evaluation of MOG II electrical infrastructure cost estimate.

⁴⁴ These reasons were communicated in the final report on the results of the TF PEZ public consultation organised by Elia:

1. The negative position of Ofgem regarding Nautilus on account of the lack of clarity of the situation of the grid on the Belgian side;
2. The negative impact of single node on Belgian industry due to the impossibility of concluding PPAs (between bidding zones)
3. The confirmation of the technical challenges associated with single node for Elia, the HVDC supplier and wind farm developers

⁴⁵ Source: Elia presentation Princess Elisabeth Island (PEI) Update (MOG II) of 12 November 2024 as explained at the hearing of the Energy, Environment and Climate Committee in the Federal Parliament

5.2. EVOLUTION OF DETAILED BUDGET PER PACKAGE

77. The costs of the five packages have evolved differently over successive budget estimates, as illustrated in Figure 6. This figure shows the evolution of the MOG II budget per package for the period October 2021 to November 2024.

Figure 6 - Evolution of budget per package ⁴⁶

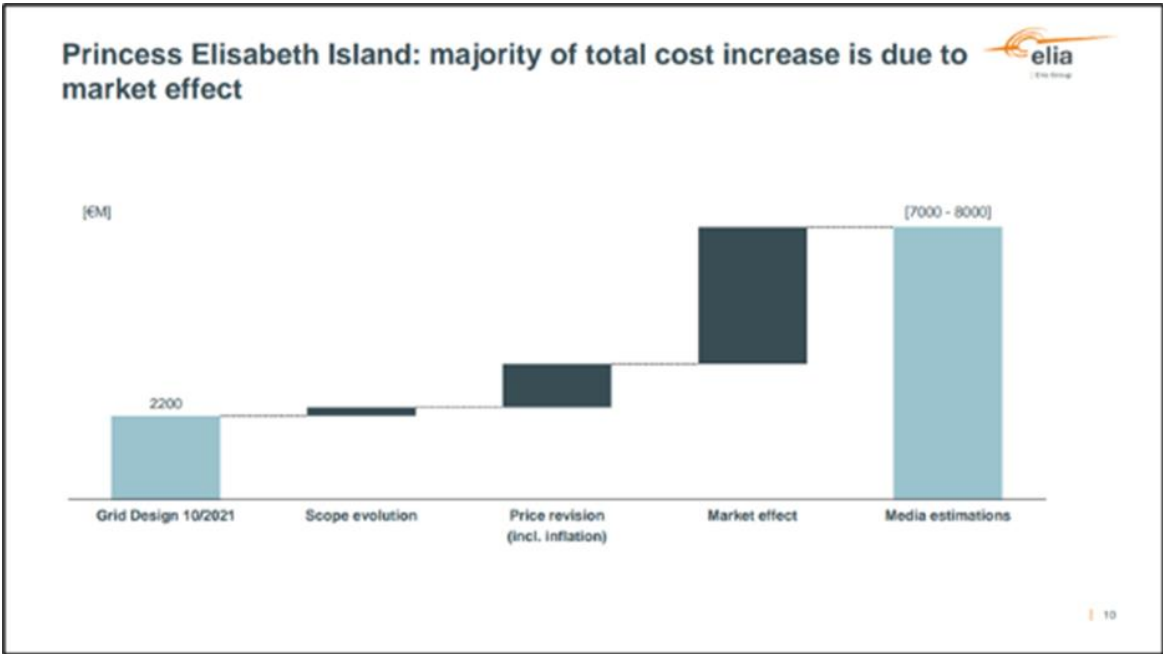
[CONFIDENTIAL]

Significant cost and/or budget increases can be observed on each of the five distinct packages of the MOG II project. The most significant cost increases between the 2021 budget and the November 2024 budget can be found in AC substations (+ 450), DC substations ([CONFIDENTIAL]) and DC cables ([CONFIDENTIAL]). The DC substations alone represent an increase of [CONFIDENTIAL].

It is important to note that at the time this report was drafted, the contracts for the artificial island packages, AC substations and AC cables had been awarded. However, the contracts for the HVDC substations and DC cables are still to be awarded. These are also the packages for which the largest cost increase has been observed in absolute terms.

78. Both during bilateral discussions with the CREG and during the hearing at the Energy, Environment and Climate Committee of the Federal Parliament, Elia identified market effects that were beyond its control as the main cause of cost increases (figure 7). Elia therefore appears to indicate that the changes to the grid design decided after October 2021 had no material impact on the identified budget increases. The scope of the assignment entrusted by the CREG to DNV, as described in the next chapter, is to assess this.

Figure 7 - Reasons for cost increases identified by Elia ⁴⁷



⁴⁶ [CONFIDENTIAL]

⁴⁷ Source: Elia presentation Princess Elisabeth Island (PEI) Update (MOG II) of 12 November 2024 as explained at the hearing of the Energy, Environment and Climate Committee in the Federal Parliament

5.3. EXTERNAL INDEPENDENT AUDIT

79. On 19 September 2024, the Minister for Energy asked Elia and the CREG in a letter⁴⁸ to carry out a detailed analysis of the costs for MOG II. The minister referred to an external independent audit to allow an objective evaluation of the cost structure. The CREG subsequently decided to solicit an external consultant to carry out the independent audit.

80. On 14 November 2024, a kick-off meeting was organised with a team from DNV Netherlands BV (hereafter DNV), the external consultants appointed by the CREG, to go through the practical organisation of the assignment and to agree on the scope of the assignment.

81. On 15 and 22 November 2024, the CREG organised a consultation with Elia, firstly to inform Elia of the assignment entrusted to DNV and secondly to reach a number of practical agreements with Elia on the exchange of information.

82. The assignment awarded to DNV included: supporting the CREG, by means of technical and economic expertise, in examining the costs of the MOG II project, with a special focus on the award of the DC substation contract:

- at technical level:
 - the choice of the technical design based on the grid design;
 - the difference between minimal vs. 'nice to have', and standardised vs. non-standardised technical requirements, mature vs. non-mature technology;
 - the evaluation of risk assessments (spare parts, backup systems, etc.);
 - the evaluation of availability requirements with regard to the standards used;
 - the risk mitigation measures or alternative technical options put in place to reduce the cost impact;
- at economic level:
 - the evaluation of budgets, including the risk allocation between contracting parties and the impact on unforeseen expenditure (additional costs);
 - a reasonableness analysis on the evolution of budgets;
 - the comparison of budgets with other similar infrastructures (international projects, onshore projects);
 - evolutions in the supplier markets, including trend analysis both for the past and into the future;
 - an evaluation of the timing of the project, taking into account future trend analysis in the supplier market.

83. DNV's mission included a technical study with a comparison of the grid design against the requirements of the Belgian government, the evolution of the grid design, the technical choices made and the reasons and justification for these choices. The assignment also included an economic study, with a cost comparison between the MOG II budget and that of comparable infrastructures, the reasonableness of the budget evolution, the impact of market developments on the budget and the

⁴⁸ See § 64.

procurement strategy. The main question to be answered is: what are the reasons for the significant budget increase between 2021 and 2024?

84. DNV's mission was focused on the award of the DC substation contract, partly because this is still an ongoing procedure and partly because this part of the project has seen the biggest budget increase.

85. As the technical design of the DC substations at MOG II also has an effect on the AC infrastructure of the landing point onshore, DNV also incorporated the analysis of this infrastructure into its mission.

86. To perform the assignment, DNV received information on:

- the policy and choices made regarding offshore wind generation and grid design by the Belgian government: documentation accompanying the Federal Development Plan, including CREG (A)2445 opinion of 15 September 2022;
- the initial grid design: CREG (A)2302 opinion of 26 November 2021 with confidential annex;
- the evolution of the grid design and the reasons for any modifications: documentation provided by Elia;
- the budget evolutions from the grid design in 2021, to the FID⁴⁹ in 2023 and the last known budget in the fourth quarter of 2024: documentation provided by Elia;
- the tender process and resulting bids: documentation provided by Elia.

DNV analysed the information received and verified and compared (= benchmarked) this information with both technical and economic information available within their own databases. Several Q&A sessions were organised by the CREG with Elia and DNV. These were held on 2, 9 and 17 December 2024.

87. Weekly consultations on the work performed by DNV were organised with the CREG. These follow-up meetings were held on 21 and 29 November, 6, 12 and 19 December 2024 and 8, 16 and 23 January 2025.

5.4. DETAILED ANALYSIS OF THE RESULTS - CAUSES OF BUDGET INCREASES

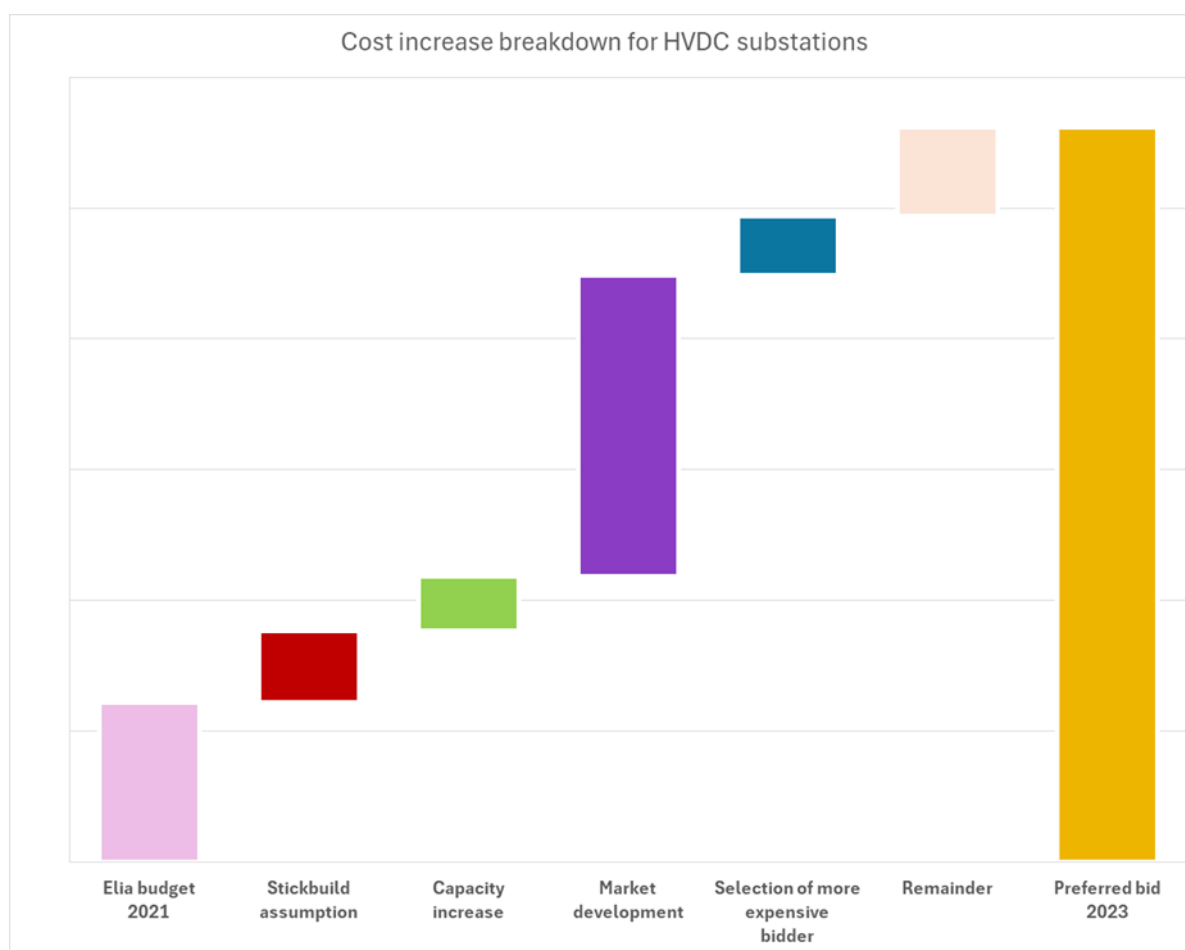
88. DNV's analysis clearly shows that in addition to market effects, various other factors played a role in the budget increases of the MOG II project. These factors and their cost-increasing effect can be linked to specific (technical) choices made by Elia. Furthermore, the choices made by Elia also amplified the cost increases due to market effects. Below is a summary of the findings from DNV's full report for each aspect individually, with additional remarks provided by the CREG. DNV's findings are based on cost estimates that can be considered best estimates based on available information.

⁴⁹ see footnote 21.

5.4.1. HVDC System (substations and cables)

89. For the HVDC substations, DNV made a breakdown of the budget increases, by benchmarking against a number of international projects. The period analysed runs from October 2021, i.e. the evaluation of the grid design, to the available bid at the end of 2023 from the preferred bidder, with whom Elia is currently negotiating. The result of this benchmarking is summarised in the following graph. The various factors explaining the budget increases are analysed one after the other below.

Figure 8 - Reasons for budget increase identified by DNV [NON-CONFIDENTIAL VERSION]⁵⁰



The X-axis of the graph above shows the following aspects:

- Elia budget 2021 = Elia's initial budget estimate for HVDC substations as included in the file to evaluate the grid design in October 2021;
- Stick-build assumption = effect of the premature assumption by Elia that onshore construction techniques (= stick-built) could be used on an energy island;
- Capacity increase = effect of capacity increase from 1.4 to 2 GW DC;

⁵⁰ Source: DNV Report, Technical and economic evaluation of MOG II electrical infrastructure cost estimate.

- Market development = market effects for HVDC substations;
- Selection of more expensive bidder = effect of the selection of the most expensive bid within the procurement process and the choice of the preferred bidder;
- Remainder = the remaining difference for which no specific allocation could be made to arrive at the amount proposed by the preferred bidder;
- Preferred bid 2023 = the amount at the end of 2023 proposed in the procurement process by the preferred bidder.

5.4.1.1. Market effects: Assessment of market capacity for HVDC substations

90. DNV acknowledges that the market effects cited by Elia had a significant impact on the project. Indeed, the market is currently tight⁵¹ meaning that suppliers have strong negotiating power that resulted in an extremely sharp price increase over a relatively short period of time. However, DNV's analysis also shows that a number of design choices made by Elia made the impact of these market effects worse. Whether these design choices were sufficiently justified is discussed in next parts of the report.

91. DNV concludes that future developments in the HVDC market are uncertain. On the demand side, global, and more specifically European, demand is expected to remain high due to pressure from commitments made to meet renewable energy targets. On the supply side, there are various trends. On the one hand, several factors such as commodity price inflation, limited competition, high entry barriers for potential new suppliers and geopolitical tensions put pressure on the supply. On the other hand, current suppliers are investing in additional production capacity to meet the increased demand. However, these investments require significant time. Furthermore, it is difficult to estimate whether these investments will fully cover the additional demand. *In globo*, therefore, no definitive answer can be provided as to which direction prices will move. The CREG therefore judges on the basis of this information that it is highly unlikely that prices will fall dramatically in the medium term. An additional element is that there are no guarantee regarding the lead time if the project is postponed, Elia would in that case have less control over the completion date for the project.

92. Based on these elements, it can be concluded that postponing the DC part would not reduce costs for society. Regarding any benefits generated by the project, which would also be lost due to a postponement, we refer to the detailed analysis in Chapter 6 of this report.

⁵¹ The market for HVDC projects is highly concentrated on the supply side, with only three suppliers (consortia) currently able to deliver projects of this kind.

5.4.1.2. Grid design assumption: energy island makes it possible to work with onshore technologies

Table 3 - Variant analysis of the grid design (2021) – Total CAPEX with separate indication of other costs/risks⁵²

Overview CAPEX per variant					
Variant 1: 5 * 700 MW platforms	M EUR	Variant 2: AC/HVDC island	M EUR	Variant 3: AC/HVDC platforms	M EUR
Other (PM, insurance, risk, overhead, ...)	240	Other (PM, insurance, risk, overhead, ...)	235	Other (PM, insurance, risk, overhead, ...)	260
CAPEX	2220	CAPEX	2205	CAPEX	2440

93. The table above shows a comparison of the costs for the three variants that were taken into account in the evaluation of the grid design. Variant 2, which features the construction of an island, gives the lowest CAPEX and the lowest amount for other costs, including risks, in this variant analysis⁵³. The difference with the total investment for variant 1 is very small and amounts to less than 1 % (≈ 0.7 %).

94. When the grid design was evaluated in 2021, it was assumed, as shown in Elia's documentation⁵⁴, that if an artificial island was to be built, the same design principles and construction techniques as on land (stick-built) could be used. Assumptions on the cost of onshore infrastructure and more specifically for AC and DC substations were therefore used for the offshore infrastructure of variant 2 in the comparison between the different conceptual variants.

95. With the stick-built approach, an enclosure is first built, and equipment is then installed component by component. The components and the parts are transported to the island by regular supply ships. After installation and commissioning, tests are carried out to validate correct installation before the substation is commissioned.

96. During further development of the MOG II project, it soon became clear to Elia that the stick-built assumption would entail significant additional time and quality risks, and that a modular or topside approach⁵⁵ was necessary, as is more common in offshore projects. Modular or topside production is significantly more expensive than stick-built. Moreover, the transport and installation ships needed for modules or topside are scarce and much more expensive compared to ships that can be used for a stick-built option.

97. Elia conducted a more comprehensive analysis and comparison between stick-built, topside and modules in 2022⁵⁶, after the decision for an island had already been made. Given the innovative and unique nature of installations on an artificial energy island, DNV concludes that it would have been reasonable to expect that Elia carried out this more comprehensive comparison before evaluating the grid design, so that the additional costs and risks (to the extent known) could have been accurately taken into account in the variants analysis. The CREG regrets that this information, both the analysis and the potential impact on the variants analysis, was not proactively communicated to the CREG and the relevant authorities. This could have potentially resolved a number of questions and concerns about the cost-effectiveness of the energy island.

⁵² A more detailed table and comments can be found under § 40. CREG opinion on the grid design (A)2302.

⁵³ This variant analysis does not take into account the impact (and therefore costs) on the inter array cables of the different variants (see §§40 and 47 above), nor the costs of decommissioning the island.

⁵⁴ Source: Document "Modular Offshore Grid - Phase 2 project for the extension of the offshore transmission grid", Elia, 15 October 2021

⁵⁵ Stick-built is a traditional method where each component is built on site. All materials are transported to the site and the structure is built piece by piece.

In 'topside' or modules, also known as prefabricated or off-site construction, construction takes place in a factory-controlled environment. The modules are then transported and installed on site (topside) or connected together (modules).

⁵⁶ Source: Elia documents: *Concept Study Report HVDC Construction Feasibility Assessment* – MOG2, Elia 25/03/2022; CBA AC Topside vs Modules – MOG2, Elia, 02/06/2022

98. The book *How Big Things Get Done* puts forward *Think Slow, Act Fast* as one of the key principles to successfully realise projects. Realising projects successfully means: within the set budget and timing.

Research by Prof Bent Flyvbjerg shows that planning is the most important phase in any project. Planning involves developing, researching, analysing, testing and detailing the original vision of the project in such a way that there is a reliable road map of how the project can be brought to a successful conclusion, so that the project can be realised smoothly and quickly.

Think Slow, Act Fast

What's the worst that can happen during planning? Maybe your whiteboard is accidentally erased? What's the worst that can happen during delivery? Almost any nightmare you can imagine can happen – and has happened – during delivery. You want to limit your exposure to this. You do it by taking all the time necessary to create a detailed, tested plan. Planning is relatively cheap and safe; delivering is expensive and dangerous. Good planning boosts the odds of a quick, effective delivery, keeping the window on risk small and closing as soon as possible.

How Big Things Get Done, p. 188

99. Elia's assumption that building an island would allow to use onshore techniques, but above all the fact that it did not carry out a comprehensive comparison between the stick-built, modular and topside approaches until after the grid design had been evaluated, suggests that Elia wanted to 'Think Fast', risking delays in the rest of the process (Act Slower).

100. According to DNV, Elia's premature assumption about the possibility of stick-built when the grid design was evaluated led to an underestimate of the budget of an offshore DC substation⁵⁷ of around € 272 million in 2021. For this, DNV compared the cost price for the construction of a 1.4 GW offshore conversion station using benchmark projects known in 2021 with the budget as prepared by Elia. The CREG concludes that if Elia had not based its grid design on an inaccurate assumption that underestimated the cost of building the electricity infrastructure on the island, variant 1, with five 700-MW platforms, might have been selected as the most cost-efficient solution in 2021.

101. Besides identifying a significant impact on the budget, DNV concludes that the original justification for choosing an offshore energy island, that it was more cost-effective as onshore construction techniques could be applied to the island, should be once again called into question.

102. In a separate chapter of the report, DNV incorporates a number of alternative grid designs. As a starting point for these alternatives, DNV uses the connection of 3.5 GW of offshore wind. The infrastructure of the MOG II design proposed by Elia is much more extensive than what is usually the case in offshore wind energy projects. In some of these alternatives, DNV, while using innovative, already operational technical solutions, also proposes infrastructure at different voltage levels than the ones currently used by Elia onshore. Among other things, these alternatives mean there is less need for infrastructure, which can result in cost savings and also potentially reduce the island area (or even its total elimination)⁵⁸.

⁵⁷ Elia's premature assumption regarding the possibility of stick-built when the grid design was evaluated also had an effect on the observed cost increase for AC substations

⁵⁸ The alternative designs proposed by DNV are:

- 1) A split node design with a direct connection of the HVDC converter
- 2) operation at a higher or non-standard voltage level in AC
- 3) an HVDC connection only
- 4) an HVDC connection only, with a lower voltage.

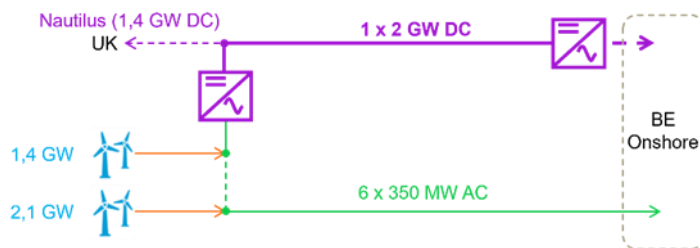
However, these designs do not always meet all the constraints and functionalities originally required for MOG II.

5.4.1.3. Increasing HVDC capacity from 1.4 GW to 2 GW

103. The grid design as presented on 15 October 2021 envisages an HVDC connection of 1.4 GW and 525 kV. This capacity is the same as the Nautilus interconnector and the maximum wind capacity to be connected in addition to the planned 2.1 GW of AC power.

However, in early 2022, Elia examined the possibility of increasing the capacity of the DC system from 1.4 GW to 2 GW, as illustrated in Figure 9. At that time, Elia believed that the capacity increase from 1.4 to 2 GW would result in an additional limited cost of € 76 million⁵⁹ for the DC stations. For the DC cables, Elia estimated that the cost increase would be limited to € 56 million.

Figure 9 - Illustration of the system with two 2 GW DC inverters and DC cables (in bold)



104. At the technical meetings between Elia, DNV and the CREG, Elia justified this development on the basis of the following elements:

- 1) Standardisation: the main reason is that the suppliers indicated that the standard capacity for their 525 kV offshore equipment was 2 GW. [CONFIDENTIAL]. Elia therefore believed that a capacity of 2 GW was necessary to organise a competitive bidding process.
- 2) Increased socio-economic welfare: Elia estimated that a capacity of 2 GW would enhance socio-economic welfare, mainly due to the possibility of increasing imports through Nautilus when wind production is not nominal.
- 3) Increased redundancy: a capacity of 2 GW increases redundancy in the event of a technical failure of an element of the AC infrastructure, by providing additional transmission capacity.

105. At the end of 2022, the decision to purchase a system with a capacity of 2 GW was taken by Elia, but they did not proactively inform the CREG nor any other government body, and no regulation stipulates such a capacity. This capacity was then explicitly stated in the technical specifications of the project, so suppliers therefore did not have the option to propose a 1.4 GW system.

⁵⁹ Amount based on Elia's responses to DNV.

106. The CREG notes that the increase in costs related to the upgrade to a capacity of 2 GW is estimated by DNV to be €211 million, only for DC stations and based on available information in 2022, i.e. without taking into account any market effect. The CREG highlights the fact that DNV's estimate is almost three times higher than Elia's. The increase in the cost of DC cables must be added on top of this. This is estimated by DNV to be €65 million, an amount similar to Elia's estimate. The CREG also points out that Elia did not request a cost estimate for a 1.4 GW system from suppliers during the project, although one supplier had indicated that it could provide this capacity. The CREG believes that, before taking a decision on increasing the HVDC capacity from 1.4 GW to 2 GW, it would have been necessary to objectify the cost difference based on information from the suppliers (rather than based on an internal estimate from Elia).

107. DNV's report also indicates that the additional costs of upgrading to a capacity of 2 GW also increased due to significant market effects. In total, these additional costs of increasing the capacity of the DC system are estimated by DNV to be €620 million for the DC stations and €65 million for the cables.

108. Regarding the additional benefits of the capacity increase from 1.4 to 2 GW, the CREG believes that the arguments put forward by Elia should be nuanced:

- 1) **Standardisation:** Elia imposed the 2 GW capacity in the specifications, so that suppliers did not have the option to propose a 1.4 GW capacity. DNV states that it probably would have been possible for suppliers to adapt their standard to a 1.4 GW capacity by reducing the capacity of certain components. The CREG points out that Elia acknowledges in this regard that one of the suppliers stated that it could propose a 1.4 GW solution. The CREG notes that keeping the HVDC capacity at 1.4 GW could have led to a significant reduction in the cost of the DC part of the project.
- 2) **Increased socio-economic welfare:** the CREG notes that Elia did indeed make simulations in 2022 which showed that increasing the capacity to 2 GW would have a positive effect on socio-economic welfare. This is mainly due to increased use of the Nautilus interconnector.

However, the CREG notes that this increased use is primarily when the system is operated in a split node configuration (i.e. with separate AC and DC export circuits, as described in §§ 116 to 123). Indeed, the CREG estimates that the additional capacity in a single node configuration can only be used to a limited extent due to the limited capacity of the onshore grid (mainly limited to 3.5 GW) and the maximum capacity of Nautilus (1.4 GW). Therefore, if the system is operated as a single node at the end of the support period for the first two offshore farms (currently set at 20 years), the benefits of increasing capacity to 2 GW will be significantly reduced.

Finally, the CREG notes that Elia's analyses of the socio-economic welfare⁶⁰ were not re-evaluated following the cost increase.

⁶⁰ For further explanations on the calculations of socio-economic welfare, see Chapter 6.

- 3) Increased redundancy: the CREG acknowledges that the upgrade to 2 GW adds 600 MW of capacity that can be used as redundant capacity if one or more AC components fails. However, it is not certain that this capacity can be used if the system operates in split node, as is currently planned. Elia also did not submit a study quantifying the cost benefits of this additional redundancy. Furthermore, DNV stated that it was not aware of currently operated wind farms with redundant capacity higher than peak generation.

109. In conclusion, the CREG notes that increasing capacity brings certain benefits (more standardisation for manufacturers, enhanced socio-economic welfare and more redundancy), but entails a significant additional cost. The CREG highlights the lack of quantitative data to objectify the additional costs compared to the benefits. The additional cost cannot be accurately evaluated since the suppliers were not asked to estimate the cost of a 1.4 GW solution. In addition, DNV and the CREG did not receive the analyses of the impact on socio-economic welfare nor the availability studies. The analyses of the CREG and DNV regarding the additional costs and benefits are different from Elia's. In addition, the CREG notes that Elia, when deciding on the expansion from 1.4 GW to 2 GW DC capacity, did not make analyses on possible cost reductions on other infrastructure parts of the MOG II. While the CREG acknowledges that the challenging market conditions may have influenced this decision, it regrets that Elia did not explicitly and proactively communicate the capacity increase in the draft specifications to the CREG. Indeed, the CREG notes that the capacity increase represents a significant change to the approved grid design, but it was not proactively and formally communicated to the authorities.

5.4.1.4. HVDC substations tendered through preferred bidder

110. In the procurement process for the HVDC substations, Elia opted for a procedure in which a preferred bidder would be used.⁶¹

111. Elia received two bids which, in addition to different technological solutions, each with their intrinsic risks, proposed different prices. The following table shows DNV's main comments on both bids.

⁶¹ Source: MOG2 HVDC converters - Procurement Strategy addendum 3 , Elia, 3 May 2023

Table 4 - Analysis of bids for HVDC substations ⁶²

	Bid 1	Bid 2 (= preferred bidder)
Design	• One large topside structure that includes all offshore DC equipment • Approach in accordance with offshore platform installations	• Multiple modules with different parts of the offshore DC equipment • Modulation of HVDC converters is unique and the first known application on MOG II
Offshore transportation	• SPMT (Self-propelled modular transport) • Specialized and exceptionally equipped ships needed for transportation	• SPMT (Self-propelled modular transport) • Transportation of modules possible with more standardized ships
Price	€ [CONFIDENTIAL]	€ [CONFIDENTIAL]
Price difference in relation to the preferred bidder	€ -[CONFIDENTIAL]	
Risk	• Delays because of specialized transportation by ship, for which there may not be sufficient capacity • The passing down of the risks is mainly attributed to Elia.	• New design that has never been applied before • Greater risk to further cost increase in realization of the project
Evaluation of supplier's performances	• Track record in realization of 525 kV VSC (Voltage Source Converters) projects	• No experience with 525kV VSC (Voltage Source Converters) projects • No modifications demanded to Elia's technical specifications, possible underestimation of risks with possible higher costs in the future

112. [CONFIDENTIAL]. It should be noted in this regard that there are currently only three suppliers in a position to submit bids for projects of this kind, one of which had already withdrawn at the start of the process.

On the basis of the available information at this moment, the choice for bid 2 as preferred bidder has a cost-increasing effect of € [CONFIDENTIAL].

⁶² Summary table based on the contents of *DNV Report, Technical and economic evaluation of MOG II electrical infrastructure cost estimate*.

Elia used the following evaluation criteria in the tender process:

Figure 10 - Evaluation criteria for bids for HVDC substation ⁶³

ID	Award Criteria	Weight
1	Mandatory HSE Criteria	Pass/Fail
2	Commercial Evaluation	55%
2.1	Monetary Evaluation	20%
2.2	Price Revision Evaluation	15%
2.2	Legal Evaluation	20%
3	Technical Evaluation	45%
Total		100%

The price criterion (Monetary Evaluation and Price Revision Evaluation) was not marked by Elia as a primary criterion for comparing bids.

In the end, Elia chose to only continue negotiations with the preferred bidder. The main problem here is that the highest bidder was chosen, [CONFIDENTIAL]. DNV raises a number of concerns about the weighting given to the evaluation criteria for comparing bids.

The CREG notes that in such a large-scale project, Elia does not consider cost reduction as a primary criterion.

113. Based on the analysis of the preferred bidder's bid and the way in which Elia laid down the technical specifications (Employer Requirements) in the tender documents, DNV identified a number of additional points that could result in additional costs during the project implementation phase, such as: future price changes as a fully developed design is not yet available, higher electrical losses as there are no specifications, possible deviations from technical specifications and modifications to the modules.

DNV did not estimate these potential additional costs. However, it is important to take them into account during the further implementation phase of the project.

114. The procurement process was carried out using a preferred bidder procedure. It is unclear to DNV why an Innovation Partnership was not used, as this was the choice of the German TSOs⁶⁴. Elia argued that working through an Innovation Partnership would have extended the set timing for the tender and that there was not enough innovation in the HVDC offshore energy island to justify this process. The innovation part would have been limited to the actual design and transport of the structures to the island.

DNV concludes that the project is indeed sufficiently innovative, as it is the world's first offshore HVDC energy island, including the infrastructure's modular design.

The CREG adds that the project proposed by Elia is so innovative at this stage that Elia is not currently sure that the equipment envisaged to eventually achieve single node operation will be able to work.

⁶³ Source: Elia document 2.5.2.2.1 Shortlisting Note_MOG2 HVDC Substations (1), p.2

⁶⁴ Source: <https://www.tennet.eu/de-en/news/energy-future-made-europe-multiterminal-hubs-important-building-block-realising-climate>

5.4.1.5. Residual amount

115. Between DNV's observations and the bid of the preferred bidder for the HVDC substations, a residual amount of € [CONFIDENTIAL] remains. This difference can be attributed to several factors, including:

- uncertainty in the cost estimates used by DNV;
- the additional risk premium passed on by suppliers, as this is an island project which has never been implemented before and for which, consequently, no information is available;
- the impact of additional market developments not fully incorporated in DNV's benchmarking, as this benchmarking was based on data from the past, while suppliers look to the future for their pricing.

DNV's benchmarking is also based on point-to-point HVDC systems and does not include infrastructure costs for the possibility of multi-terminal extension with future interconnectors envisaged in the MOG II design. These costs were not separately identified but include at least the costs for: the HVDC switchgear, the design, the design's development and engineering studies and replica equipment for control and security.

5.4.2. HVAC System (substations and cables)

5.4.2.1. Operation of the system in single node or split node

116. The difference between a single node and a split node configuration is described in detail at the beginning of this chapter. As a reminder, in a single node configuration, all wind farms, the DC converter and the AC export cables are electrically connected offshore. On the other hand, in split node configuration, the AC and DC export systems are not connected. Single node configuration allows the power to be exported to be distributed flexibly between AC and DC systems. . However, this configuration requires more electrical equipment and is therefore more expensive.

117. In the grid design file of 15 October 2021, no choice is made regarding the configuration in single or split node. Elia does not express a preference for one or the other configuration in that file, and indicates that this choice will be the subject of further studies⁶⁵.

⁶⁵ Source: Document "Modular Offshore Grid - Phase 2 project for the extension of the offshore transmission grid", Elia, 15 October 2021 p. 30-31:

Two possible options are shown in the figure above, namely a direct transformation from the 66 kV substation to the HVDC conversion station or transformation via a 220 kV substation (a combination of the 2 options as shown for illustration on the figure is not possible). Some advantages of the 220 kV substation (if coupled with the other 220 kV substations on the island to connect wind farms via AC) are increased flexibility in terms of grid operation, more reliability and an identical connection method for all connected farms. However, this results in an additional cost and more space taken up, which can vary significantly depending on whether the infrastructure is located on an artificial island or on a platform (see section 4.4). This optimisation (and possibly others) are the subject of further studies. The final single-wire scheme of the complete system will be refined at a later stage depending on the choice of scenario, the total capacity of the wind farms, the size of the plots, etc.

Elia justifies studying a single node system with the following arguments:

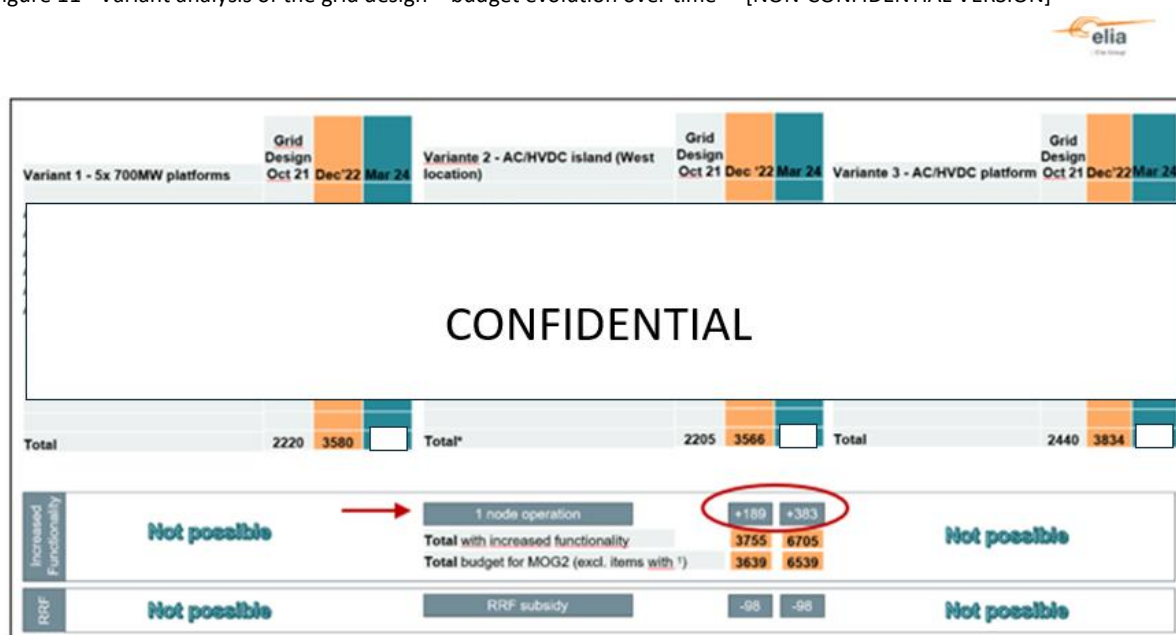
- 1) The possibility to make optimal use of different connections to the coast. Indeed, a single node system makes it possible to use AC cables as much as possible when wind production is not nominal, thereby maximising the DC capacity available for interconnectors. This enhances socio-economic welfare.
- 2) Increased redundancy if the AC or HVDC cable fails. In such cases, it is possible to export part of the wind production through the other system.
- 3) Easier commissioning of the last wind farm (because it is possible to decouple the planning of the farm from the planning of the HVDC infrastructure).

However, Elia underlines major technical challenges⁶⁶ and challenges in terms of market integration related to the single node configuration.

118. At the end of 2022, Elia confirmed its ambition to operate the system in single node. In early 2023, Elia included the required electrical equipment in the FID budget, which entailed an estimated additional cost of €189 million (see figure 11). This decision was taken without seeking an opinion from the CREG or the government, and the regulation does not stipulate this.

119. In 2024, however, Elia decided to operate the system in split node, while the necessary equipment for operation in single node would be installed (it had already been ordered). This would allow the system to be operated in single node after the support period for the wind farms, provided the technical challenges identified by Elia were resolved. Elia estimated the additional cost to be €383 million (see Figure 11).

Figure 11 - Variant analysis of the grid design – budget evolution over time ⁶⁷ [NON-CONFIDENTIAL VERSION]

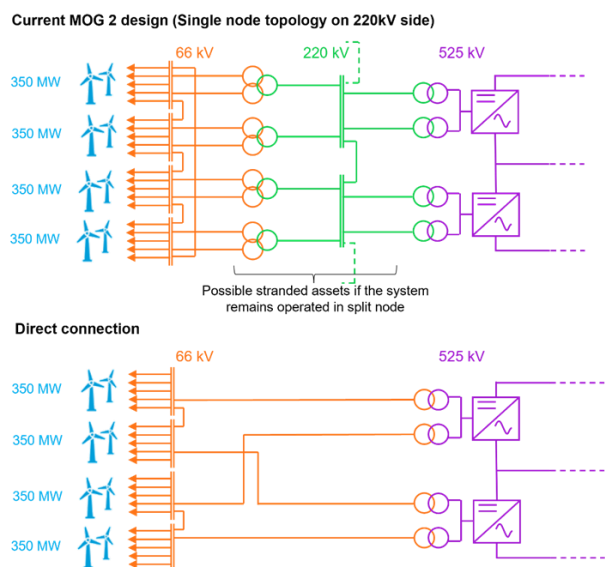


⁶⁶ For example, the technical limitations of AC export cables or the impact on electronic control systems on the capacity of the HVDC converter and wind turbines.

⁶⁷ Source: Elia document Context & Historical overview_V1, p.10.

120. DNV and the CREG point out that in the vast majority of connection projects for wind energy, the preferred option is a direct connection between offshore wind turbine cables and the transformers at the interface of the HVDC converter. The difference between the topology currently planned by Elia and a direct connection is illustrated in figure 12. This configuration with direct connection minimises the need for electrical equipment to be installed offshore, while ensuring a generally acceptable availability. The configuration is used in many international offshore projects (e.g. TenneT in the Netherlands and Germany, RTE in France and Amprion in Germany). The configuration currently provided by Elia allows a single AC node operation, but DNV stresses that this single node configuration is internationally not very well-known as the benefits generally do not outweigh the costs. In the grid design document, Elia also proposes a direct connection as an option and states that the choice will be made at a later date, based on the optimisation studies. DNV requested this analysis but did not receive it.

Figure 12 - Difference between the current MOG II design (with single node topology) and a direct connection (figure adapted from the DNV report). The 220 kV infrastructure (green) is primarily necessary for single node operation⁶⁸



121. The costs estimated by DNV for this additional equipment is €44 to 64 million, plus transport and installation costs, as well as the impact on the occupied surface on the island. Elia's updated budget of €383 million therefore seems coherent. DNV also points out that the single node topology requires a higher system reliability. Indeed, in the event of a serious outage, the risk in single node configuration is that all transmission capacity of the system fails, i.e. potentially 3.5 GW in the case of the MOG II. DNV highlights the use of 'double busbars', while 'single busbars' are usually used offshore to reduce equipment costs. The CREG therefore notes that, compared to other designs, the topology chosen by Elia requires a high degree of redundancy, but this means higher costs.

122. The CREG notes that the choice of a single node topology was made after the initial grid design. Despite the significant additional cost, this choice was not explicitly communicated to the CREG or the government. The CREG acknowledges that a single node operation could have had a positive impact on socio-economic welfare and reliability, but regrets⁶⁹ that Elia did not provide quantitative data when

⁶⁸ Source: DNV Report, Technical and economic evaluation of MOG II electrical infrastructure cost estimate.

⁶⁹ CREG opinion on grid design (A)2302 (freely translated): "The CREG understands from the grid design dossier that an energy island potentially offers benefits at the level of available capacity in N-1 situations and has a positive impact on social economic welfare, but notes that these arguments are not sufficiently and quantitatively elaborated. Consequently, the CREG cannot evaluate the estimated impact of these potential benefits."

studying this solution. This would have made it possible to objectively weigh up the additional costs against the benefits.

123. Finally, the CREG notes that with the current choice to initially operate the system in split node configuration, the additional electrical equipment will be of little use. If the system were to continue to operate in split node for technical or market reasons, this additional equipment will never be used.

5.4.2.2. Oversizing and configuration of AC substation onshore

124. The onshore substation is not formally part of the MOG II project, but it is part of the Ventilus project. However, from a grid design perspective, it is linked to the design of the MOG II's offshore AC system. DNV therefore assessed this design as well. Based on the information provided, two aspects of avoidable costs were identified in the design of the onshore substation: oversizing of transformer capacity and the substation's configuration. DNV has not received any information demonstrating that Elia considered different substation designs, nor that Elia made comparisons to determine which substation provided the best cost-benefit ratio for the full life span of the infrastructure.

125. The onshore AC substation has five transformers, each of 600 MVA, with a total capacity of 3 GVA. This is 900 MVA or 43 % more than the required 2.1 GW export capacity of the AC export cables. According to Elia, the aim of the oversizing was mainly to achieve N-1 redundancy⁷⁰.

126. DNV is not aware of any other AC export connections for offshore wind farms that have onshore redundant transformer capacity. The fact that Elia has not⁷¹ analysed whether the benefits of redundant transformer capacity outweigh the costs prompts the CREG to conclude once again that Elia does not consider cost reduction as a primary criterion in project development.

127. The onshore switchgear is a triple busbar single breaker configuration. In this configuration, the MOG II system can continue to operate at nominal capacity if a busbar failure occurs while another busbar is out of service. According to DNV, this configuration is very reliable and offers excellent operational flexibility, which is generally standard in meshed onshore transmission networks.

DNV believes that this type of substation configuration is highly atypical, since it is assumed, for offshore wind energy export connections, that the benefits of such a configuration do not outweigh the significantly higher costs.

Also for this component, Elia did not analyse whether the benefits of the chosen switch configuration outweigh the costs.

⁷⁰ It is important to note that the N-1 redundancy criterion was not applied by Elia to the capacity of the AC export cable.

Redundancy: the term redundancy is used to describe systems that have built-in back-up components that immediately take over tasks if the primary components fail (for example, as provided for in Elia's onshore mesh network)

⁷¹ Or whether such analysis was not proactively shared with the CREG and other authorities

128. The CREG would like to add that the identified oversizing and configuration of the onshore AC substation resulted in a significant expansion of the infrastructure and therefore added to the cost increase⁷², confirming the lack of a cost reduction reflex on Elia's side.

5.4.3. Spare parts and extension of the infrastructure

129. In the context of DNV's audit, Elia provided a list of spare parts and extension capacities for the infrastructure. Spare parts make maintenance easier in the event of system failure, and neither DNV nor the CREG call these into question. Most of the elements that enable the future expansion of the system are described by DNV as an industry standard practice or a plausible future application. However, DNV states that there is a high likelihood that four 220 kV switchgear bays would remain unused. DNV estimates the purchase cost of these switchgear bays at €10 million, to which and additional cost for transport and installation could be added.

5.4.4. Surface area of the island

130. The CREG notes that between the grid design proposed in 2021 and the FID budget in 2023, the total area of the island also increased from 5 to 6 ha. Between these two budget estimates, the estimated cost of the island increased from €504 M to €729 M. Although the impact of the increase in the surface area of the island is not explicit in the budget, it likely had a significant impact on costs. The required surface area of the island is strongly correlated to the infrastructure that needs to be installed, and efficiency gains and cost savings may possibly be realised here as well, amongst others should the DC part in fine not be built.

5.4.5. Future evolutions of the MOG II budget

131. In its analysis of the HVDC system, DNV identified a number of supplementary issues that could lead to additional costs during the project implementation phase.

132. A further increase in the investment amounts (CAPEX) potentially also has a cost-increasing effect on operational expenditure (OPEX). A large share of the OPEX is linked to insurance costs, which in turn have a strong connection with the underlying value of the insured infrastructure.

133. [CONFIDENTIAL].

Figure 13 - MOG II budget November 2024

[CONFIDENTIAL]

⁷² These increases are also partly due to market effects, which are again magnified as infrastructure is added

This amount represents Elia's estimate of possible future price effects linked to contracted prices that include an indexation formula or a cost+ principle.

Elia identifies three different price formulas in the MOG II contracts:

Lump sum = fixed price contracted with the supplier and there are no changes in the future.

Indexed = price set by the supplier based on a formula linked to indexation parameters (e.g. raw material prices), whereby price variations are possible.

Cost+ = supplier charges its actual cost including margin at the time of delivery.

134. The current split between lump sum, indexed and cost+ is 13 % lump sum, 78 % indexed and 8 % cost+⁷³. This breakdown applies to the elements laid down by Elia in EPCI (Engineering, Procurement, Construction and Installation) contracts, which account for approximately 80 % of the total budget.

135. The real total cost of the MOG II will not be known until the entire project is completed. Only then will it also become clear whether the price revision reserve of [CONFIDENTIAL] estimated by Elia was sufficient, but also here is a risk of further cost increases for the MOG II project.

136. At the hearing of 27 November 2024 in the Energy, Environment and Climate Committee of the Federal Parliament, Elia stated that the figures disclosed at the time of the evaluation of the grid design, in October 2021, were only an initial rough estimate and cannot be considered as a budget for the MOG II project.

Among other things, the book *How Big Things Get Done* underlines the danger of strategic misrepresentation.

Strategic misrepresentation

..., the tendency to deliberately and systematically distort or misstate information for strategic purposes. If you want to win a contract or get a project approved, superficial planning is handy because it glosses over major challenges, which keeps the estimated cost and time down, which wins contracts and gets projects approved. But as certain as the law of gravity, challenges ignored during planning will eventually boomerang back as delays and cost overruns during delivery. By then the project will be too far along to turn back. Getting to the point of no return is the real goal of strategic misrepresentation. It is politics, resulting in failure by design.

How Big Things Get Done, p. 26-27

The question that arises here is whether, in addition to failing to inform the relevant authorities fully and in good time, Elia also engaged in strategic misrepresentation in this project.

⁷³Source: budgetary information from Elia submitted to the CREG and included in the presentation to Elia's board of directors on 28 November 2024

5.5. CONCLUSION

137. The budget increases of the MOG II system can be attributed to:

- the market developments for HVDC and AC infrastructure over the period 2022-2024, due to global shortage in the supply chain, and in particular for offshore applications. These market developments resulted in limited supply, with higher bids, less flexibility and increased risk for Elia;
- Elia's focus on ensuring a grid design with high availability and operational flexibility led to assumptions and design choices that increased the cost base compared to typical offshore transmission designs. Examples of this include operation with a single AC node, the HVDC connection's higher capacity and the inclusion of redundant and spare equipment;
- the urge to stick to a strict timing for delivery as a result of which unverified assumptions led to premature design choices. The main factors in this regard are the following: the assumption that the offshore HVDC and AC substations could be built on the island via onshore techniques (stick-built method) and the technical and regulatory feasibility of using a single AC node.

138. The external audit conducted by DNV shows that market developments did have a significant impact on the evolution of the costs of the MOG II, but that a number of elements linked to choices made by Elia also played a role in the significant cost increases.

Table 5 – Causes and quantification of cost increases due to (technical) choices by Elia

	CAUSES AND QUANTIFICATION OF COST INCREASES DUE TO (TECHNICAL) CHOICES BY ELIA	Increase compared to the grid design budget of October 2021
1	HVDC-System	
	Assumption application onshore technology on an energy island	€ [CONFIDENTIAL]
	DC Capacity increase 1,4GW to 2 GW	€ [CONFIDENTIAL] DC-substations + € [CONFIDENTIAL] DC- cables
	Surcharge preferred bidder	€ [CONFIDENTIAL]
	Future budget increases – additional costs project implementation phase	Figures currently not available
2	HVAC-System	
	Assumption application onshore technology on an energy island	Figures currently not available
	Single node versus split node	€ [CONFIDENTIAL]
	Overdimensioning and configuration AC onshore substation (part of Ventilus project)	Figures currently not available
3	Spare parts and spare infrastructure	Figures currently not available
4	Island surface (Spare space)	Figures currently not available
5	Future budget increases – price revision reserve sufficient?	Figures currently not available
	TOTAL	At the least € 1.570 million

Based on the detailed analysis carried out by the CREG and DNV, €1.570 billion cost-increasing effects have been identified that can be directly linked to specific (technical) choices made by Elia.

In addition to the above-mentioned figures, a number of possible additional budget increases were identified that could (significantly) increase the above-mentioned amount. The external audit also reveals that an overestimation of the transformer capacity and the configuration of the Ventilus onshore substation may also have led to cost increases.

The CREG regrets that significant changes were made to the conceptual design without the authorities being explicitly informed. In the future, the CREG would like Elia to proactively communicate significant changes to the scope of major projects to the CREG, as well as to the authorities, in order to ensure continuous monitoring.

If after an evaluation, Elia comes to the conclusion that the tendering for and further development of the DC infrastructure on the energy island should be abandoned, the question could be raised whether the surface of the island could be reduced. To finance the construction of the island, Elia takes an RFF grant of €100 million into account. Payment of this grant is conditional on the island being completed within strict deadlines. Elia is currently facing a possible delay in the construction process of the island, which may jeopardise the above-mentioned grant. An adjustment (reduction) of the island's surface area would seem to be worth considering in this context.

⁷⁴ Zie § 88

In the context of its tariff competence, the CREG will continue to monitor and assess the reasonableness of the costs incurred for the MOG II, and will systematically verify whether these costs should be charged to Belgian grid users.

6. COMPARISON OF COSTS AND BENEFITS OF POSSIBLE VARIANTS FOR MOG II

139. This chapter evaluates the costs and benefits of the current MOG II design and other possible variants presented by Elia.

140. The following cost-benefit analyses (hereafter, “CBA analyses”) aim to assess the social, economic and environmental impacts of an infrastructure project of general interest. The analyses therefore make it possible to compare the costs of this infrastructure project with its expected benefits, assess the social merits, inform the decision-making authorities and support political decision-making. When several designs are possible for the same infrastructure project, a cost-benefit analysis makes it possible to compare these alternatives and decide which alternative offers the most benefits in relation to the costs.

141. The cost-benefit analyses therefore support the investment decision.

6.1. GENERAL CONSIDERATIONS

142. In the first instance, it should be recalled that the development and implementation of grid infrastructure projects is the transmission system operator’s core activity, in accordance with the applicable legal framework and taking into account the planned budgets and timetables. Cost-benefit analyses should generally be carried out prior to any decision to implement the project. Since Elia has the most complete and up-to-date information on the costs and benefits of an infrastructure project, it is the most appropriate party to carry out these cost-benefit analyses. Moreover, Elia has knowledge of both the Belgian and international electricity systems, the experience and the model to perform the necessary simulations to estimate the impact of a project on market prices in different countries and on the contribution to socio-economic welfare and the impact on society as a whole.

143. The CREG on the other hand does not have the resources (either in terms of data or simulation models and the associated financial resources, human resources and experience) to reproduce the simulations made by Elia, and therefore it cannot validate their results. Based on a comparison of Elia's simulations - in which Elia was asked additional specific information questions - the CREG aimed to challenge and assess the consistency between the simulations and the plausibility of the results.

144. Cost-benefit analyses - in the context of infrastructure works of general interest - should also be considered from the perspective of society. From Elia's perspective, all investments in the transmission network are remunerated with a profit margin via tariffs to the extent that the costs are not deemed unreasonable: the regulated asset base (RAB) increases with every investment. The business case for additional investments is therefore always positive from the perspective of the regulated transmission system operator, and independent from the added social value of these investments.

145. In the context of the federal development plan, Elia performs cost-benefit analyses for the proposed, undecided projects for which the contribution to social economic welfare (SEW) can be calculated, namely for projects in the "horizontal system": interconnectors and the 380 kV grid.

146. Elia's methodology for performing cost-benefit analyses in the context of investment projects included in the federal development plan is set out in section 1.4.5. of the 2024-2034 Federal Development Plan. In this regard, Elia applies a methodology based on the methodology proposed in the third CBA guideline of ENTSO-E⁷⁵, the European Network of Transmission System Operators for Electricity. This third CBA guideline was submitted for approval by ENTSO-E to the European Commission on 27 October 2022, pursuant to Regulation (EU)347/2013 of 17 April 2013⁷⁶. Cost-benefit analyses for investments compare the costs of the project with the benefits over the lifetime of the project.

147. The benefits include the positive (or negative) impact created by a project. For the cost-benefit analyses in the 2024-2034 federal development plan, Elia only takes into account a limited number of indicators from the third CBA guideline in order to calculate the benefits, in particular those underlined in the list below:

- B1: Social Economic Welfare⁷⁷;
- B2: Additional societal benefit due to CO₂ variation;
- B3: RES integration⁷⁸;
- B4: Non-CO₂ emissions;
- B5: Grid losses;
- B6: Security of supply - Adequacy;
- B7: Security of supply - Flexibility;
- B8: Security of supply - Stability.

⁷⁵ Source: https://eepublicdownloads.blob.core.windows.net/public-cdn-container/tyndp-documents/CBA/221019-3rd%20CBA_Guidelines%20.pdf

⁷⁶ Source: <https://www.entsoe.eu/news/2024/04/09/entso-e-publishes-the-final-guideline-for-cost-benefit-analysis-of-grid-development-projects/>

⁷⁷ The Social Economic Welfare reflects the impact, of commercial exchanges (imports & exports) through the interconnector in question, on the producer surplus (PS), consumer surplus (CS) and congestion rents (CR): $SWE = PS + CS + CR$. In its development plan, Elia defines these concepts as follows (section 1.4.5.1. of the 2024-2034 Federal Development Plan):

The consumer surplus is defined as the difference between the wholesale electricity price and the maximum price that market participants are willing to pay for it.

The producer surplus is defined as the difference between the revenues (wholesale electricity price times traded volume) earned by producers and the variable production costs.

The congestion rent is defined as the price difference between an importing and exporting bidding zone multiplied by the energy exchanged (on an hourly basis).

⁷⁸ RES: Renewable energy sources

The following costs are included and taken into account by Elia (in line with those in the third CBA guideline):

- C1: Capital cost (CAPEX) expressed in €;
- C2: Operating expenditure and maintenance costs (OPEX) expressed in €/year.

148. To calculate the SEW, Elia uses the "Antares" model. First of all, it should be noted that any model simulates a simplified reality, with several limitations, and whereby reality is imitated by simulating certain scenarios. Even if the model can give a close approximation of reality, the results of the simulations (outputs) of future scenarios depend on the data and hypotheses used (inputs).

Since these are future scenarios, the underlying hypotheses are by definition uncertain (moreover, this uncertainty rises as the relevant time horizon broadens), but their impact on the results can be decisive. Consequently, the results of the simulations should be interpreted in the light of and within the limits of the hypotheses used.

149. The main factors influencing the quantification of the benefits, and the SEW in particular, are the following:

- the evolution of the generation facilities and the energy mix in Belgium and neighbouring countries;
- the evolution of the grid infrastructure and the chosen reference grid;
- the evolution of electricity demand (which is strongly influenced by the rate at which electrification emerges);
- the evolution of CO₂ costs and fuel prices;
- the evolution of market coupling rules within the EU and with the UK.

Within the electricity system, any change made to it (e.g. earlier or later commissioning or decommissioning of an interconnector, whether or not a project will be realised) has an impact on the electricity flows and therefore also on the SEW.

150. It should be noted that the MOG II design was decided on before the most recent federal development plan was approved⁷⁹. For the MOG II project, whose estimated costs are evolving at an unprecedented rate, no quantitative cost-benefit analysis was carried out by Elia in 2021 in preparation for the decision on the grid design, but only a cost comparison (budgeted costs) combined with a qualitative benefit analysis between different variants⁸⁰. In the 2024-2034 Federal Development Plan, the MOG II project was considered as a confirmed investment and consequently no CBA was made for the MOG II project. It was only on 13 December 2024⁸¹ that Elia submitted to the CREG the results of simulations of the social economic welfare analysis for MOG II.

⁷⁹ The grid design for MOG II was approved at first reading on 23 December 2021 (see §41).

⁸⁰ See § 39. § 40 discusses Opinion (A)2302 and the CREG's main remarks on the variants analysis.

⁸¹ A SEW analysis was made for energy hub in the 2024-2034 federal development plan. This does not give an overview of the welfare generated by the MOG II, as the energy hub has a more extensive scope.

6.2. CONSIDERATIONS IN CHOOSING THE VARIANTS TO BE FURTHER EVALUATED

151. In determining the different variants to be considered, as requested at the hearing of 27 November 2024 in the Energy, Environment and Climate Committee, it is first necessary to determine the limits within which adjustments or modifications to the MOG II are possible. The CREG believes that the following elements, among others, are important in this consideration:

- the decision of the Ministerial Council of 15 October 2021 to develop 3.5 GW of offshore wind farms in the Princess Elisabeth Zone by 2030 (see § 38);
- the decision on the grid design of the MOG II project (see § 41);
- the contracts already concluded by Elia (for the construction of the island and the purchase of the AC components) (and any claims for compensation in the event of changes or breach of these contracts);
- [CONFIDENTIAL].

152. Based on the planning submitted by Elia, the CREG notes that the development of 3.5 GW of offshore wind by 2030 can, strictly speaking, no longer be respected. The commissioning of the DC infrastructure and therefore the connection of the third parcel of the Princess Elisabeth Zone is not expected before the end of 2033 at the earliest (see § 157). Based on the available information, it appears that, both in the current MOG II design and in possible variants (see section 6.3.1), the initially envisaged time frame (2030) cannot be respected.

If the decisions taken, with the exception of the timing, are regarded as fixed, then no variants are possible apart from delaying (part of) the implementation of the MOG II project.

In order to comply with the request to examine variants, a greater or lesser degree of deviation from previously made decisions on the development of offshore wind turbine capacity in the Princess Elisabeth Zone and on the MOG II design must be allowed. Any alternative grid design therefore requires at least a review of the decisions already taken by the government.

153. With regard to the contracts already concluded by Elia (for the construction of the island and the purchase of the AC components), no breach or modification of the scope is envisaged. Indeed, this would probably give rise to damage claims and delays of the connection of the first parcel of the wind farms in the Princess Elisabeth Zone. In particular with regard to the construction of an island versus platforms, the current cost of construction with platforms is not known, meaning that an accurate comparison is impossible.

154. [CONFIDENTIAL].

The variant to fully connect a capacity of 3.5 GW offshore wind turbines in AC would require the installation of 4 additional AC cables⁸² (i.e. 10 instead of 6). [CONFIDENTIAL]. Consequently, this option will not be analysed further within the scope of this report.

⁸² The planned 220-kV AC cables each have a capacity of 350 MW. To bring 2.1 GW ashore via AC, 6 AC cables are therefore necessary. If the full 3.5 GW has to be brought ashore via AC, a total of 10 AC cables are needed. In the envisaged configuration where the final 1.4 GW is to be brought ashore via DC, 6 AC cables and one DC cable system are needed.

6.3. FIGURES PRESENTED BY ELIA AND COMMENTS BY THE CREG

155. At various times, Elia has provided information on the benefits, either through publications (federal development plan), hearings at the Energy, Environment and Climate Committee or directly to the CREG in response to questions it had asked.

Table 6 - Overview of dates of recent information exchange on SEW and variants. Source: CREG

Date	
8/10/2024	Slides and complementary note
12/11/2024	Energy Committee hearing
18/11/2024	Meeting on CBA Nautilus
25/11/2024	Elia's response to the CREG's questionnaire dated 14/11/2024
27/11/2024	Energy Committee hearing
13/12/2024	Presentation of variants to the CREG
21/12/2024	Partial response from Elia to the CREG's questionnaire dated 10/12/2024 and 16/12/2024
08/01/2025	Partial response from Elia to the CREG's questionnaire dated 10/12/2024 and 16/12/2024

The CREG notes that, over time, an evolution and nuance became apparent in the documents and figures presented by Elia, during the hearings in the Chamber Committee (12 November 2024 and 27 November 2024) on the one hand and following the most recent meetings with the CREG on the SEW and cost-benefit analyses on the other hand.

The CREG also notes an evolution in Elia's willingness to consider alternatives to the current MOG II project (see §§ 60, 61 and 67).

On 13 December 2024, after six months of information exchange and meetings, Elia provided the CREG with results based on new simulations, while the figures communicated previously (including during the hearings) were estimates based on deductions from the available results of simulations carried out in the context of the Federal Development Plan and the Adequacy & Flexibility Study⁸³.

The CREG believes that discussing all the information initially presented by Elia, which has been updated by more recent provided information, would lead to confusion and therefore has no added value. The CREG therefore limits itself to a discussion of the most recently communicated results and information (dated 13 December 2024). The information of 13 December 2024 supported by the files sent on 8 January 2025 provide more detailed information, an annual calculation (instead of only the results for a limited number of key years) and a better understanding of Elia's working method. The latter information enables the CREG to respond with a quantitative underpinning.

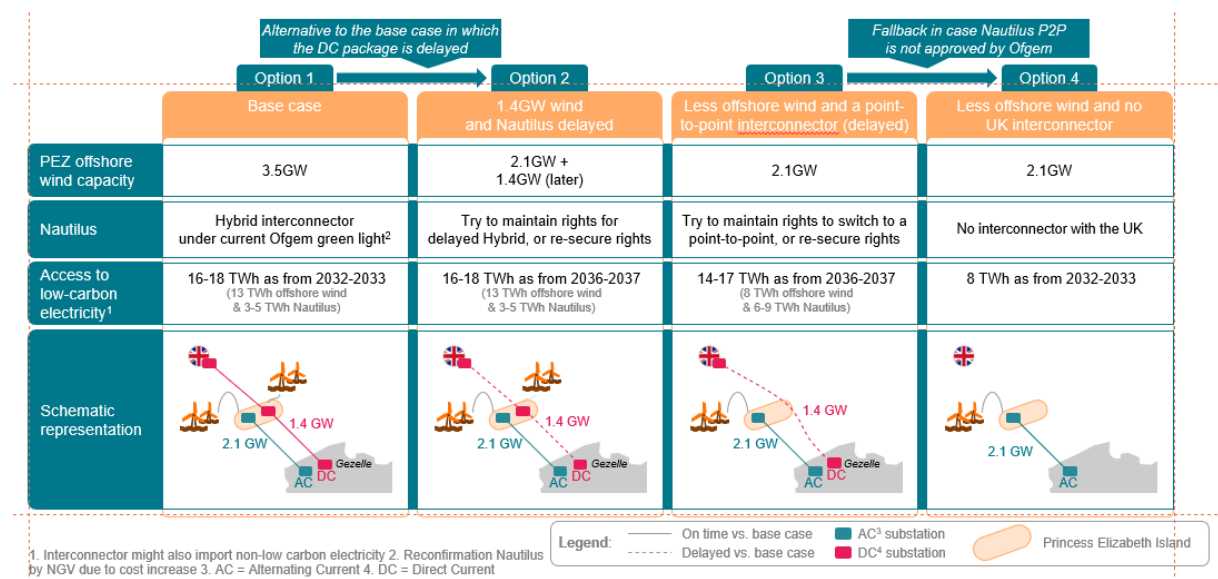
⁸³ Source: https://issuu.com/eliagroup/docs/adequacy- en_flexibiliteitsstudie_voor_belgi_202?fr=sZDlmMDYxOTUwMTY

6.3.1. Cost-benefit and variant analysis dated 13 December 2024

156. During a meeting between Elia and the CREG on 13 December 2024, Elia provided an update on the CAPEX evolution for the MOG II project and Nautilus, and compared the basic MOG II design with a number of alternatives. In order to compare the alternatives, Elia made new simulations to calculate the impact on socio-economic welfare. These results should therefore be considered as Elia's best available estimates.

157. The options, as proposed by Elia, are shown in Figure 14 and further explained in table 7.

Figure 14 - **Overview of the variants**; source: ELIA - slide 10 of presentation dated 13/12/2024 to the CREG



Based on the information provided by Elia on 8 January 2025 regarding the NPV calculation, the CREG has determined the commissioning dates in the following table. These dates all deviate, including option 1 or the basic scenario, from the assumption of 3.5 GW offshore wind connected by 2030.

Table 7 - Overview of variants and commissioning dates; source: CREG based on information from Elia dated 13/12/2024 and 8/01/2025

Option	Description	Commissioning
Option 1: basic scenario	This scenario corresponds to the current design of the MOG II project, consisting of the connection of 3.5 GW of offshore wind power in the Princess Elisabeth Zone, combined with a 1.4 GW hybrid interconnector with the UK (Nautilus).	Indienstnameschema Optie 1
Option 2: delay of 1.4 GW offshore and Nautilus	Option 2 has the same design as option 1, but is different in that the implementation of the DC part will be realised with a 3- year-delay. Specifically, this means that both the Nautilus interconnector and the last parcel of offshore wind turbines in the Princess Elisabeth Zone (1,225 to 1,400 MW) will not be operational until 2036.	Indienstnameschema Optie 2
Option 3: 2.1 GW offshore and 1.4 GW point-to-point Nautilus	In option 3, the capacity of installed wind power in the Princess Elisabeth Zone is limited to 2.1 GW and fully connected in AC, through the six AC cables also envisaged in the basic scenario. In this option, it is decided to implement Nautilus as a point-to-point interconnector of 1.4 GW, with a 4-year-delay compared to the basic scenario.	Indienstnameschema Optie 3
Option 4: 2.1 GW offshore	In Option 4, only 2.1 GW of offshore wind capacity is connected with the mainland and no additional interconnector with the UK is considered.	Indienstnameschema Optie 4

158. The CREG notes that an alternative option, based on option 4, where no additional interconnector with the United Kingdom is envisaged, could be to install two additional AC cables in order to bring the capacity of the Princess Elisabeth Zone up to approximately 2.8 GW. However, the impact of "replacing" the DC cables with AC cables also needs to be examined in this regard, without increasing the total number of cables at the onshore landing point, both in terms of technical and cost impact and [CONFIDENTIAL].

159. The table below summarises the investment costs (CAPEX) for each of the above options.

Table 8 - Overview of investment costs, € m (CAPEX) of various options; source: information Elia dated 8/01/2025

Mio €	Option 1	Option 2	Option 3	Option 4
Island	[CONFIDENTIAL]	[CONFIDENTIAL]	[CONFIDENTIAL]	[CONFIDENTIAL]
AC scope	[CONFIDENTIAL]	[CONFIDENTIAL]	[CONFIDENTIAL]	[CONFIDENTIAL]
DC scope	[CONFIDENTIAL]	[CONFIDENTIAL]	-	-
Total MOG II	[CONFIDENTIAL: between € 7 and 8 billion]	[CONFIDENTIAL: between € 7 and 8 billion]	[CONFIDENTIAL]	[CONFIDENTIAL]
Nautilus (50 %)	[CONFIDENTIAL]	[CONFIDENTIAL]	[CONFIDENTIAL]	-
Total MOG II + Nautilus	[CONFIDENTIAL: between € 8.5 and 9.5 billion]	[CONFIDENTIAL: between € 8.5 and 9.5 billion]	[CONFIDENTIAL: between € 5 and 6 billion]	[CONFIDENTIAL: between € 3 and 4 billion]

The CAPEX costs are Elia's best estimates at the end of 2024 and may vary in the future; especially for option 2, the CAPEX is uncertain, as in this option the DC part will only be contracted at a later date at currently unknown prices (higher or lower than the current envisaged cost).

As regards the cost of the island, it can be ascertained that this investment cost remains constant for all options. It can be noted that for options 3 and 4, the island could be smaller in size (because installing a DC part is no longer necessary) which entails a lower cost (see section 5.4.4).

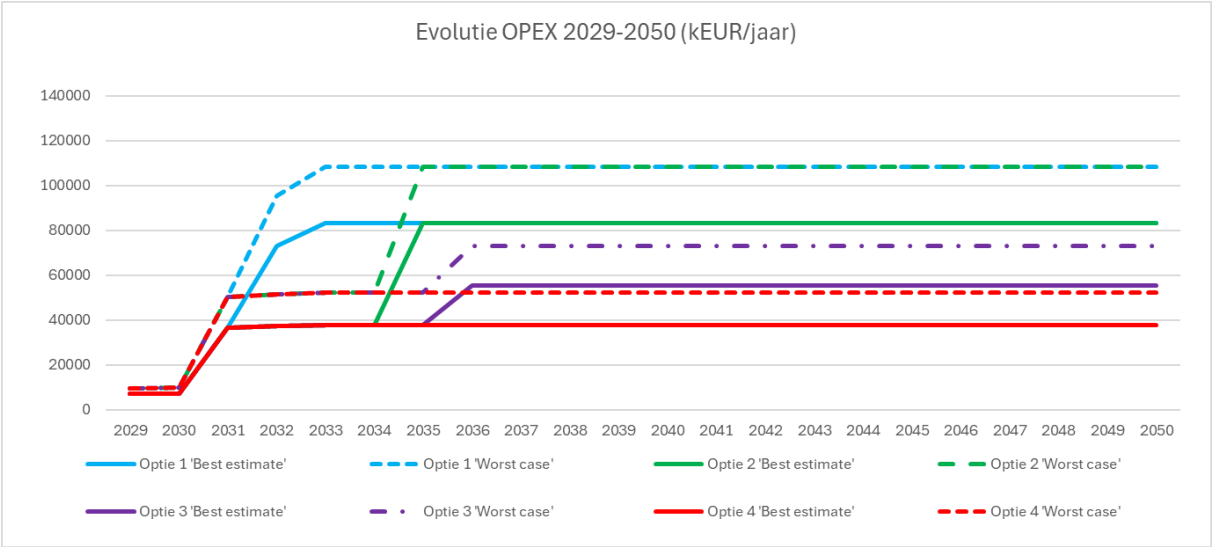
160. For the operational costs (OPEX), Elia proposes a low (best estimate) and a high variant (worst case). The table and figure below show the evolution of the OPEX of the four options in both variants.

Table 9 - Overview of the OPEX in k€ /y of different options in two variants; source: information Elia dated 8/01/2025

[CONFIDENTIAL]

The OPEX for 2034 was added to the table to illustrate the difference between Option 1 and Option 2. The graph gives a better visual representation of the evolution of the OPEX.

Figure 15 - Comparison of OPEX for different options in both variants; source: CREG based on information from Elia dated 8/01/2025



The CREG notes that the evolution of the OPEX does not quite correspond to the commissioning dates of MOG II and Nautilus. The operational costs increase in some cases at the time of planned commissioning and sometimes prior to the commissioning.

The CREG also notes that the OPEX is considered constant from full commissioning onwards, although these costs may still increase.

It can also be established that the DC part on the island has a significant impact on OPEX (compare options 1 and 2 with options 3 and 4 in this regard).

161. Elia analysed and compared the four options for various indicators. The results of this analysis are presented below.

Table 10 - Evaluation of the four options; source: Elia - slide 11 of presentation of 13/12/2024 to the CREG

	Option 1 Base case	Option 2 1.4GW wind and Nautilus delayed	Option 3 Less offshore wind and a point-to-point interconnector (delayed)	Option 4 Less offshore wind and no UK interconnector
Low-carbon supply?	16-18 TWh as of '32-'33 (75% domestic)	16-18 TWh as of '36-'37 (75% domestic)	14-17 TWh as of '36-'37 (50% domestic)	8 TWh as of '32-'33 (100% domestic)
In line with Ministerial Decree	Yes	Yes	No	No
Benefits by 2040?	€7-7.5/MWh 800MW adequacy contribution as from 2032	€7-7.5/MWh 800MW adequacy contribution as from 2036	€6-7/MWh 700MW adequacy contribution as from 2036	€3-4/MWh 800MW adequacy contribution as from 2036
Cost by 2040?	~€4.9/MWh (~€7.7/MWh at the peak)	~€5.1/MWh ¹ (~€7.8/MWh at the peak)	~€3.5/MWh (~€4.7/MWh at the peak)	~€2.9/MWh (~€3.5/MWh at the peak)
Reputation risk towards partners?	Some, due to tariff increase	Some, to suppliers and UK	Some, to suppliers and UK	High, to suppliers and UK
Biggest risk?	Project Execution	CAPEX uncertainty DC ²	Depending on Ofgem approval	Negative bill impact consumers

At the meeting on 13 December 2024, Elia explained that based on this analysis, option 1 was still its preferred option.

162. In general, the CREG notes that the choice of colours in the table above is suggestive, whereby the current MOG II design (option 1) contains the most green. The CREG prefers a comparison without colours, in other words a neutral table with only figures and text, which is a more objective way of presenting the variants. Furthermore, the CREG has several specific reservations about Elia's calculations and the figures presented. These are discussed below by topic.

Indicator 1: Low Carbon Supply

163. The CREG notes that the estimate of the amount of low-carbon supply corresponds to the maximum of the potential, but in reality may be lower for several reasons.

The range of 16-18 TWh/year in options 1 and 2 consists of 13 TWh/year offshore wind energy and 3-5 TWh/year of imported energy via Nautilus⁸⁴. The range of 14-17 TWh/year in option 3 consists of 8 TWh/year offshore wind energy and 6-9 TWh/year of imported energy via Nautilus.

With regard to offshore wind energy, a capacity factor of 42.4 % is therefore taken into account for Options 1 and 2 and a capacity factor of 43.5 % for Options 3 and 4 (the higher capacity factor for Options 3 and 4 is due to the lower installed capacity and therefore fewer wake effects as parcel 3 is not built).

Firstly, the estimate of low-carbon electricity from wind turbines in the Princess Elisabeth Zone may have been too optimistic for all options. Indeed, the capacity of the future wind turbines envisaged in the three parcels of the Princess Elisabeth Zone is not fixed at 3.5 GW, but can vary between 3.15 GW and 3.5 GW⁸⁵. Given the high density in the parcels, there is a possibility that candidate project developers would rather opt for the lowest possible density in order to limit wake effects and optimise the profitability of the project. In this case, wind energy production could be up to 10 % lower than stated by Elia. Furthermore, in options 1 and 2, part of the electricity production in the Princess Elisabeth Zone will also be exported to the UK via Nautilus. When calculating the low-carbon supply in Belgium, these exports should therefore be deducted.

It should also be noted that Elia assumes that offshore wind turbines in the Princess Elisabeth Zone can inject without curtailment. If negative prices are bid in the UK, there is a risk under options 1 and 2 that Belgian wind generation will not be able to inject because of the limited capacity on the 1.4 GW DC connection due to low prices in the UK and the associated exports via Nautilus from the UK to Belgium.

⁸⁴ Energy distribution based on Elia's responses to the CREG's questions dated 21/12/2024.

⁸⁵ Source: Tender document "Coordination of preliminary studies for the Princess Elisabeth Zone offshore tender - Princess Elisabeth Zone project and site description", Table 3-1 Overview of the plots within the Princess Elisabeth Zone, with their surface area, average water depth and proposed total installed capacity. See website of the FPS Economy: https://offshore-tender-platform-pez.my.site.com/s/?language=en_US

Second, regarding the simulated imports of electricity via Nautilus, it should be noted that there is no guarantee that only low-carbon electricity is imported from the UK. The import volumes shown are also subject to significant uncertainty and are highly dependent on political preconditions regarding the market integration of Nautilus (see also §171).

Moreover, it should be noted for the sake of clarity that Elia's Nautilus simulations only take into account the net volumes of electricity imported from the UK via the interconnector.

Finally, the difference between the estimate of the volume imported via Nautilus in Options 1 and 2 and in Option 3 is explained by the different configurations of the interconnector: a hybrid configuration in Options 1 and 2 and a point-to-point configuration in Option 3. A hybrid interconnector allows both the transmission of electricity between two countries and the transmission of electricity produced by offshore wind turbines to the coast. In the case of the hybrid configuration of Nautilus, the 1.4 GW capacity of the interconnector is only partially used for electricity transmission between Belgium and the UK. On the other hand, the point-to-point configuration of Nautilus would only allow electricity transmission between the two countries, implying that the total capacity of the cable can be used to import electricity from the UK (or to export from Belgium). Import volumes via the Nautilus point-to-point configuration are therefore higher than in the hybrid configuration.

The comments above show that the values for low-carbon supply provided in Elia's table above are not transparent and are a maximalist approximation of the potential and therefore may be an overestimate.

Indicator 2: Conformity with the Ministerial decree

164. The CREG believes that in the context of comparing the options, the degree of conformity with previous policy decisions should not be a decisive element. As noted earlier (see § 152), a deviation from the existing regulatory framework is inherently linked to an examination of variants.

By using highly contrasting colours from green to red, in a table where all indicators are given the same weight, it is implicitly argued to keep the chosen design of the MOG II.

Indicators 3 & 4: Benefits and costs in 2040

165. The CREG notes firstly that the results depend on the choice of the reference year 2040, for which the results are shown, and that the costs and benefits are not evenly distributed over the years. These observations may influence the objective comparison of the different options.

166. Secondly, the CREG notes that the benefits in this table are presented in terms of wholesale electricity price falls.

The CREG notes that Elia expects a sharp drop in the wholesale price thanks to an additional interconnector with the UK, either through a hybrid configuration (options 1, 2) or through a point-to-point configuration (option 3). If Nautilus is built, the impact of 3.5 GW offshore wind (options 1, 2) instead of 2.1 GW offshore wind (option 3) on the wholesale price is relatively small.

167. As regards the impact on the average wholesale price, Elia expects a fall in the average wholesale price in 2040 of € 7 to 7.5/MWh in Option 1, of € 7 to 7.5/MWh in Option 2, of € 6 to 7/MWh in Option 3 and of € 3 to 4/MWh in Option 4. Based on these estimates by Elia, the price impact of Nautilus is more significant than the price impact of the final 1.4 GW of offshore wind.

168. As regards the costs for the end consumer, expressed in €/MWh and taking into account expected growth on the demand side, Elia estimates the impact of the various options as follows: € 4.9/MWh for Option 1; € 5.1/MWh for Option 2; € 3.5/MWh for Option 3 and € 2.9/MWh for Option 4.

169. Given the lower estimated cost for Option 3, Option 3 is therefore, based on these figures, the most favourable from a consumer perspective. However, this is without taking into account the fact that the costs and benefits of Nautilus are delayed by four years in Option 3 compared to Option 1.

170. Besides calculating the benefits as regards the wholesale price, Elia also calculated the benefits in terms of Belgian SEW for the period 2030-2040. As indicated in § 147, the SEW gives a more general picture of the benefits of a network investment than just the tariff impact⁸⁶

	Option 1	Option 2	Option 3	Option 4
SEW	€3.8 - 5.9 B	€3.3 - 5.1 B	€2.6 - 3.9 B	€2.1 - 3.2 B

Source: Elia (slide 24 of presentation from 13 December 2024)

These benefits, expressed in terms of SEW, need to be weighed against the investment costs to compare the Net Present Value of the 4 options based on SEW.

171. In general, it should be emphasised that the presented benefits of the options with an interconnector to the UK assume the most favourable preconditions for the market integration of this interconnector. This means that the following assumptions are made 1) there is an implicit market coupling between the UK and the EU, 2) there are no import restrictions due to the CBAM regulation⁸⁷ and 3) the effective implementation of the 70% capacity requirement for market coupling within the European Union⁸⁸. The CREG points out that none of these conditions are currently fulfilled. Stakeholders at Belgian and European level, including Elia and NemoLink, are taking initiatives to put these issues on the political agenda and to resolve the above-mentioned problems. If these conditions are not met in the future either, the benefits for Nautilus will be (possibly significantly) lower than currently assumed in the simulations (SEW reduction of more than 50 %). This will jeopardise the profitability of Nautilus (and EU-UK interconnectors in general). A clear, sufficiently substantiated and transparent cost-benefit analysis of the Nautilus project is essential.

172. As regards the outcomes in terms of security of supply, the 100 MW difference between Options 1 and 2 (800 MW contribution) and Option 3 (700 MW contribution) is due to a combination of the following two effects:

- the reduction of offshore wind capacity by 1.4 GW, representing a reduction in security of supply of approximately 150 MW;
- the configuration of the interconnector - 'point-to-point' versus hybrid. Elia applies qualitative reasoning to quantify this impact. During periods of scarcity, offshore wind generation is limited⁸⁹. This means that, in the hybrid configuration of Nautilus, the 1.4 GW cable between the Princess Elisabeth Zone and the Belgian coast is available (but

⁸⁶It should be noted that SEW is only one indicator for assessing an investment in the grid. As explained in §147 (chapter 6.1 General considerations), there are other indicators to measure the benefits in terms of reducing CO₂ emissions, integrating renewables, reducing grid losses or also the contribution to security of supply and physical security of installations.

⁸⁷ The CREG refers here, among others, to the Afry study "*EU CBAM impact study focused on electricity imports from Great Britain*", summary report of 6 March 2024, commissioned by Elia and Nemo Link, among others.

⁸⁸ See article 16(8) of EU Regulation 2019/914

⁸⁹A shortfall situation occurs when electricity production is insufficient to meet demand. Such a situation may arise in particular when renewable energy production is limited due to poor wind conditions and/or limited sunshine. In the context of security of supply (CRM) simulations, Elia assumes that the contribution of offshore wind energy is around 10% during periods of scarcity.

not completely) to import electricity from the UK. Elia then believes that Nautilus in hybrid configuration makes a significant contribution to security of supply at times of scarcity compared to offshore capacity. In the 'point-to-point' configuration, the contribution to security of supply is larger because the full cable capacity can be used to import electricity from the UK. However, Elia reasons that Nautilus' contribution to security of supply in point-to-point configuration is insufficient to compensate for the lower offshore wind capacity, resulting in a drop of 100 MW between options 1 and 2 and option 3.

The CREG further notes that the contribution to security of supply in option 4 contains a material error, confirmed by Elia on 13 January 2024 (the figure of 800 MW should be 200 MW). The limited contribution to security of supply in option 4 is explained by the limited contribution of offshore wind to security of supply during the periods of scarcity simulated by Elia.

Taking into account the above comments, the CREG believes that the comparison made by Elia of the various options in terms of their contribution to security of supply is probably a fairly accurate reflection.

Indicator 5: Risk of reputational damage vis-à-vis third parties

173. The assessment of the risk of reputational damage is considered from Elia's perspective. The CREG notes that concerns about Elia's reputation are considered larger with regard to foreign suppliers than with regard to Belgian electricity users who will, basically, bear the cost of the investment.

All variants have a cost much higher than the cost estimated in 2021. As far as they are considered not unreasonable, these costs are borne by electricity consumers through tariffs. It could therefore be argued that the reputational damage in Option 1 is higher than in Option 4, given that the budget overrun in Option 1 is much larger than in Option 4.

Moreover, the CREG doubts whether the reputational damage vis-à-vis the suppliers or the UK is indeed as high as suggested by Elia. If the estimated costs more than triple compared to the initial estimate, it seems obvious to the CREG that projects are called into question, as they would be if these suppliers were faced with similar cost increases themselves⁹⁰. To the extent that the cost increases are due to supplier shortages on the market, the CREG believes that Elia only incurs very limited reputational damage vis-à-vis suppliers. Finally, the CREG believes that the reputational damage vis-à-vis the UK is also fairly limited, given the current special circumstances regarding the cost increases.

174. Given the potentially subjective interpretation of reputational risk, the CREG proposes disregarding this element.

⁹⁰ For this, the CREG also refers to an article in De Tijd of 17 January 2025, "Elia in de problemen met Amerikaanse expansie". A project (in which Elia participates via EnergyRe Giga) in the US was halted because costs had risen from €3.5 billion to €5.2 billion and the subsidies were not enough. As there is no limit on passing on costs through transmission tariffs, Elia has not experienced lower profits despite the sharp rise in costs.

Indicator 6: Biggest risk

175. To effectively assess the risk of option 3, contact should be made with Ofgem.

176. Based on the concerns mentioned above, the CREG believes that when the different options are evaluated, it is mainly the impact on RES objectives (indicator 1) and the impact on costs and benefits (indicators 3 & 4) that are decisive, taking into account a consideration of the risks (indicator 6).

6.3.2. Net present value

177. To compare the impact of the costs and benefits of the four options for the Belgian end consumer, taking into account the differences in timing of implementation and the time lag between costs and benefits, Elia calculated the net present value (NPV), seen from the consumers' perspective, for each of the four options. As both benefits and costs are highly dependent on the assumed electricity demand, Elia calculated the NPV for three scenarios with different evolution of electricity demand over the time horizon 2025-2040.

6.3.2.1. Elia's assumptions and results

178. The variants for the evolution of annual electricity demand (expressed in TWh/year), namely low electricity demand, medium electricity demand and high electricity demand, are summarised in the following table for the years 2025, 2035 and 2040.

Table 11 - Variants for the evolution of electricity demand; source: information from Elia dated 8/01/2025

TWh/y	2025	2035	2040
Low electricity demand	85.7	116.8	131.5
Medium electricity demand	85.7	126.5	144.8
High electricity demand	85.7	136.1	158.0

179. Besides the variants for the evolution of electricity demand, three variants ("maximum", "average" and "minimum") for the impact on the end user in terms of congestion rents and wholesale electricity price were also quantified, expressed in €/MWh, on the basis of simulations made by Elia, for the years 2025, 2035 and 2040. Table 12 shows the variants for the impact of congestion rents and Table 13 shows the variants for the impact on the wholesale electricity price, in each case for the different options.

Table 12 - Variants for the benefits of the congestion rents; source: information from Elia dated 8/01/2025

	EUR/MWh	2030	2035	2040
Maximum	Option 1	0.3	1.5	1
	Option 2	0.3	-0.5	1
	Option 3	0.3	-0.5	0
	Option 4	0.3	-0.5	-1
Average	Option 1	0.2	1	0.75
	Option 2	0.2	-0.5	0.75
	Option 3	0.2	-0.5	0
	Option 4	0.2	-0.5	-0.75
Minimum	Option 1	0	0.5	0.5
	Option 2	0	-0.5	0.5
	Option 3	0	-0.5	0
	Option 4	0	-0.5	-0.5

The values for the benefits of the congestion rents are derived from simulations made by Elia. Positive values correspond to positive benefits, negative values correspond to costs. The total benefits are calculated by multiplying the values in €/MWh by the electricity demand for the year in question.

Table 13 - Variants for the impact on the electricity price; source: information from Elia dated 8/01/2025

	EUR/MWh	2030	2035	2040
Maximum	Option 1	1.5	8	7.5
	Option 2	1.5	4	7.5
	Option 3	1.5	4	7.0
	Option 4	1.5	4	4.0
Average	Option 1	1.2	7	7.3
	Option 2	1.2	3.5	7.3
	Option 3	1.2	3.5	6.5
	Option 4	1.2	3.5	3.5
Minimum	Option 1	0.8	6	7.0
	Option 2	0.8	3	7.0
	Option 3	0.8	3	6.0
	Option 4	0.8	3	3.0

The values for the impact on the electricity price are derived from simulations made by Elia. Positive values correspond to positive benefits for the user (i.e. a reduction in the electricity price), negative values correspond to costs (i.e. an increase in the electricity price). The total benefits are calculated by multiplying the values in €/MWh by the electricity demand for the year in question.

180. The results of Elia's NPV calculations from the perspective of electricity consumers are shown in table 14.

This table shows the net present value from the perspective of Belgian consumers, i.e. in terms of benefits, only those related to the evolution of the electricity price and the costs included in the network tariffs are taken into account.

Table 14 - Overview of the NPVs for Belgian consumers of the various options; source: Elia, slide 12 of the presentation dated 13/12/2024

NPV of PEI and Nautilus from a Belgian consumer perspective¹ [€B]

		Option 1	Option 2	Option 3	Option 4	Max and min forecasts from Federaal Planbureau, ENTSO-E, Energyville and Elia blueprint³
		Base case	1.4GW wind and Nautilus delayed	Less offshore wind and point-to-point interconnector (delayed)	Less offshore wind and no UK interconnector	
Electricity consumption scenario ²	High	2.1	1.8	3.7	0.1	
	Medium	0.9	0.7	2.8	-0.3	
	Low	-0.3	-0.4	1.9	-0.7	

! Depending on re-approval of Nautilus as point-to-point interconnector by Ofgem

181. The table shows the net present values, expressed in billion €, for the different options at different levels of electricity consumption. The table shows that option 3 not only generates the highest NPV values of all options, but is also the only option that always has a positive NPV regardless of the level of electricity consumption. Elia notes, however, that if the point-to-point Nautilus interconnector could not be built, for example, if there is no agreement with Ofgem, the result would be option 4, which has mostly negative NPV values.

6.3.2.2. Remarks by the CREG

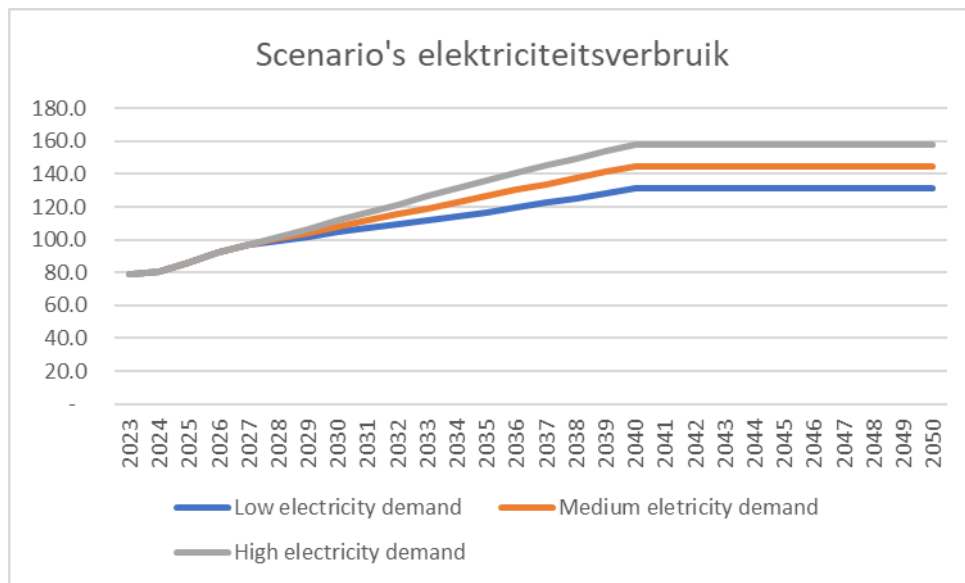
182. It should again be emphasised that the presented NPV values refer to the benefits for Belgian consumers, taking into account the impact on wholesale electricity prices and congestion rents. The impact of variation in wholesale electricity prices on the producer surplus is not considered here.

183. The calculation in Excel of the NPV values was submitted by Elia to the CREG on 8 January 2025. The CREG believes that this calculation provides a good basis for comparison, but nevertheless has some reservations.

184. A first important observation concerns the impact of electricity demand. As can also be seen from Table 14, the NPV increases as electricity consumption increases. This is explained by the fact that the NPV is the present value of the product of the net consumer impact (in €/MWh) in each year and electricity consumption in the same year. An estimate of high future electricity consumption leads to higher benefits on the one hand, which translates into a higher NPV, and to a lower cost per MWh on the other (given that the cost is covered by more MWh).

However, if the evolution of electricity consumption is lower than its current estimate, this results in a double negative effect: on the one hand, the benefits decrease (as far as they were positive), but on the other hand, the cost per MWh increases (the same investment cost will have to be covered by fewer MWh).

To reduce the risk of this double negative effect, it is recommended as a precaution to estimate the evolution of electricity consumption with moderation (and best not to overestimate it). The graph below shows the evolution of electricity consumption used by Elia in its calculations, with the addition of the recent historical values for 2023 and 2024.



Source: CREG based on figures from Elia

Electricity consumption in 2025 is estimated at 85.7 TWh, already a fair bit higher than the 2024 value published by Elia (80.5 TWh⁹¹). In the low variant, electricity consumption amounts to 131.5 TWh in 2040 (an increase of more than 63 % compared to the observed value for 2024). For the medium and high variants, this increase is 80 % (144.8 TWh) and 96 % (158 TWh) respectively.

In the past, in the context of other files (Adequacy & flexibility study, CRM, tariff files), the CREG has also highlighted Elia's tendency to possibly overestimate future electricity consumption. In the context of Elia's public consultations, other market players have also already expressed doubts as to Elia's projected evolution of electricity demand.

185. A second observation concerns the discount rate used.

A social discount rate of 4 % was used as the discount. Although the discount rate has an impact on NPV (with a lower discount rate, there are larger differences in NPV), the relative proportions are still kept. The 4 % discount rate therefore seems acceptable to the CREG and is not likely to change the conclusions.

186. Furthermore, Elia has taken into account the low variant of the OPEX (best estimate) and the average variants of the electricity price impact and impact of congestion rents in the proposed results. If the OPEX were higher than Elia's current best estimate, i.e. taking into account the high variant of the OPEX (see § 160), the NPV for all options decrease by approximately € 200 to 400 million, as illustrated in the table below.

⁹¹ Source: https://www.elia.be/en/press/2025/01/20250102_electricity-mix

Table 15 – overview of NPV with higher OPEX for the 4 options; source: calculations CREG based on Elia figures

NPV with higher OPEX (€ billion)	option 1	option 2	option 3	option 4
High demand	1.7	1.5	3.4	-0.1
Medium demand	0.5	0.4	2.5	-0.5
Low demand	-0.6	-0.7	1.6	-0.9

Option 3 then remains the only option with a positive NPV if the evolution of electricity demand is low.

187. If a pessimistic scenario is simulated, in which a high estimate of OPEX is combined with the minimum variants for the consumer impact (lowest electricity price impact and lowest congestion rents), the results are much more unfavourable, with also negative NPV with an average evolution in electricity demand as illustrated in the table below.

Table 16 – Worst case NPV for the 4 options; source: calculations of the CREG based on Elia figures

Worst Case NPV (€ billion)	option 1	option 2	option 3	option 4
High demand	0.3	0.4	2.4	-0.7
Medium demand	-0.7	-0.6	1.6	-1.1
Low demand	-1.8	-1.7	0.8	-1.4

If we exclude the scenario with the highest electricity demand for the reasons mentioned in § 184, it can be established that with pessimistic assumptions only option 3 still results in a positive NPV.

188. The various simulations made by Elia show that the Nautilus interconnector with the UK makes a significant contribution to the consumer surplus in Belgium (decrease in the electricity price in Belgium - see comments to table 10 in § 161) and is therefore highly decisive for the benefits generated in the first three options.

However, as cited in § 171, the results are strongly influenced by the assumptions on market integration used in the model. In addition, it goes without saying that the expected evolution of the price difference with the UK is decisive for the calculated benefits. Both a delay in the energy transition to be achieved on the UK side and a review of the UK bidding zone⁹² would result in fewer imports from the UK and possibly a lower price decrease for Belgium.

189. Taking into account the above-stated analyses and comments, the CREG believes it is crucial to correctly assess the risk associated with option 3, namely the possibility that Nautilus point-to-point ultimately does not go ahead and we end up with option 4, which appears to result in a negative NPV in most scenarios.

⁹² UK Department for Energy Security & Net Zero, March 2024, "Review of Electricity Market Arrangements, Second Consultation Document"

6.3.3. Cost to Belgian consumers/taxpayers

190. Besides the costs and benefits to Belgian consumers mentioned above, the CREG believes that from the perspective of Belgian consumers/taxpayers, it is also relevant to take into account the cost of subsidies for offshore wind farms (cost to society).

191. When applying the methodology for calculating the socio-economic welfare of grid infrastructure projects, as explained in section 6.1, the costs of subsidising the generation units (possibly including storage and demand response) that form the basis of the supply curves on either side of the grid infrastructure, and therefore influence the elements of socio-economic welfare, are not taken into account. This approach is acceptable in the cost-benefit analyses for interconnectors, given that whether or not the interconnector is realised does not change the award of subsidies.

As regards the benefits generated by the offshore wind farms in the Princess Elisabeth Zone, the amount of the subsidy costs⁹³ is directly linked to the realisation of the grid infrastructure. If the grid infrastructure connecting Princess Elisabeth Island to the mainland is not built, no wind turbines can be connected. Support for wind farms in the Princess Elisabeth Zone is therefore inextricably linked to the construction of the necessary grid infrastructure. Conversely, building the grid infrastructure on the island is inextricably linked to the objective of connecting and linking these wind farms to the Belgian mainland. Given that the future domain concessionaire will not necessarily be a Belgian company, possibly not even a European one, the effect of subsidies is not neutral from a Belgian perspective. The benefits derived from offshore wind energy production in the Princess Elisabeth Zone, including the impact on the electricity price due to wind energy produced in the PEZ, would not exist without the support of the wind farms. For these reasons, the CREG believes it is useful, in the context of transparent communication, to also highlight the cost of supporting offshore production in the Princess Elisabeth Zone.

192. The annual cost for MOG II (connection costs under the basic scenario - option 1) is estimated by Elia at $\approx$ € 620 million per year in 2032⁹⁴. This excludes OPEX and decommissioning costs. If these costs are also taken into account, the annual cost will increase to € 728 million for Belgian consumers, excluding the cost of Nautilus.

193. As regards the cost of supporting offshore wind farms, if a number of conservative assumptions are taken into account (strike price = € 90/MWh; electricity reference price = € 50/MWh⁹⁵), the CREG arrives at a minimum cost of € 470 to 520 M per year for 3.15 to 3.5 GW of installed capacity, respectively. For 1.925 to 2.1 GW of offshore wind capacity, the CREG arrives at a minimum cost of € 293 to 320 M. Per MWh, the support under these assumptions is € 40/MWh.

⁹³ Subsidies are the result of a political choice for a given technology, and they are necessary, depending on the evolution of the market, to actually attract the necessary investment in wind farms. On the other hand, it is the case that subsidies are borne by the consumer and partially negate any impact on the price of energy due to wind production.

⁹⁴ Slide 17 of the Elia presentation at the Energy Committee on 12 November 2024

⁹⁵ The positive impact on the wholesale price (lower price) due to the realisation of MOG II and Nautilus has a negative impact on the costs for support (which will increase as a result).

The CREG points out that this is a minimal cost under the above-mentioned assumptions, given that the effect of the correction factor or the cost of any offshore wind curtailment has not yet been taken into account.⁹⁶

In summary, the total cost to Belgian consumers (either through general funds or electricity bills) for connecting and supporting offshore wind farms is € 1.1 to 1.4 billion/year.

6.3.4. Estimates of avoided CO₂

194. Besides the cost of supporting offshore wind capacity, there are also benefits in terms of avoided CO₂ emissions. Estimates for these avoided CO₂ emissions depend on which valuation is applied per tonne of CO₂. The available figures in this regard vary too widely.

6.4. CONCLUSION

195. In Chapter 6, the CREG assesses the costs and benefits of the MOG II project, as well as possible alternative options.

First of all, the CREG stresses that conducting cost-benefit analyses of grid infrastructure projects is the responsibility of Elia which, unlike the CREG, disposes of the necessary resources and competences to do this. Given the inevitable uncertainty about future developments, the results of the simulations should always be interpreted in the light of and within the limits of the hypotheses used.

Having already identified sharp budget increases for the MOG II, the CREG asked Elia already on 20 June 2024 to examine alternative options. Only after the hearings in the Energy Committee in November 2024, on 13 December 2024, did Elia effectively present different options to the CREG.

In addition to the current MOG II grid design, three alternative options were examined by Elia (see section 6.3.1). The construction of the island and the installation of the AC infrastructure, both already contracted, are not called into question here.

The CREG has a number of comments and reservations regarding Elia's evaluation of the variants. According to the CREG, the slides presented by Elia on 13 December 2024 give an overly positive image of the current MOG II design compared to the variants. The CREG also recommends taking into account the lowest assumption for the evolution of electricity demand in the evaluation, to err on the side of caution. Moreover, the benefits presented by Elia assume the most favourable preconditions for the market integration of Nautilus, which may not materialise in practice.

In analysing the net present value calculation from the perspective of electricity consumers, the CREG concludes that, even under the more pessimistic assumptions, only Option 3 gives a positive net present value for consumers. In Option 3, the installed capacity of offshore wind in the Princess Elisabeth Zone is limited to 2.1 GW fully connected in alternating current (AC). With this option, the choice is also made to implement Nautilus as a 1.4 GW point-to-point interconnector. This third option had a total investment cost of [CONFIDENTIAL: between € 5 and 6 billion] in relation to

⁹⁶ If the grid reinforcements on the internal Belgian backbone envisaged in the 2024-2043 Federal Development Plan are realised in time (i.e. Ventilux, Boucle du Hainaut, HTLS reinforcements 380 kV-grid, etc.) Elia does not expect structural congestions that would lead to structural redispatching or RES curtailment of MOG I or MOG II. However - besides congestions - market-related RES curtailment is also possible. In a hybrid configuration of Nautilus - in the event of significantly negative prices on the UK side, the wind farm connected to Nautilus could be curtailed to give priority to imports from the UK. This possibility has not yet been confirmed or quantified by Elia.

[CONFIDENTIAL: between € 8.5 and 9.5 billion] for the current MOG II design and the Nautilus connexion.

7. EUROPEAN REGULATORS

7.1. INTRODUCTION

196. In the current context of the energy transition, a European comparison of the evolution of investment costs in electricity grid infrastructure provides an additional tool to objectify the cost evolution, especially with regard to the development of offshore grid assets.

197. A comparative evaluation makes it possible to highlight both the factors that different countries have in common, and their specific national characteristics, and draw relevant lessons.

Identifying best practices can also provide inspiration to help optimise investment expenditure and promote the efficient use of resources, in line with the CREG's recommendations in its white paper '*No transition without transmission - Financing strategic network investments as part of the energy transition*'⁹⁷.

Finally, comparing practices and costs at the European level provides a better understanding of the transparency and justification of planned investments at national level. These are essential criteria for gaining the trust of consumers and/or taxpayers, who basically have to bear the costs.

These requirements regarding transparency and the social responsibility of investments are all the more acute as the development of offshore transmission infrastructure will lead to a significant and accelerated increase in the Regulated Asset Base (RAB) of the transmission system operator, who remains the owner of these transmission assets.

198. From this perspective, the CREG, as the national regulator responsible for tariff aspects relating to grid infrastructure, has proposed setting up a specific European working group within CEER, bringing together 30 member countries⁹⁸.

199. In the shorter term, and in order to contribute as much as possible to the debate and immediate challenges at the Belgian level, the CREG recently surveyed its European counterparts, to have first indicative insights at the European level. To this end, it organised several bilateral meetings and sent out a questionnaire to European regulators adapted to the urgent circumstances of the exercise conducted in December 2024.

⁹⁷ <https://www.creg.be/fr/actualites/no-transition-without-transmission>

⁹⁸ Belgium, Bulgaria, Cyprus, Denmark, Germany, Estonia, Finland, France, Greece, Hungary, Ireland, Iceland, Italy, Croatia, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Austria, Poland, Portugal, Romania, Slovenia, Slovakia, Spain, Czech Republic, Sweden and UK.

200. With regard to the questionnaire, and in order to initially gather some informative data from the European regulators in the short term, they were asked the following questions (general and specific), following a brief description of the Belgian case:

1. *Do you face similar costs evolution regarding large investments in infrastructure assets, especially in the offshore sector?*
2. *If yes:*
 - 2.1. *How big is the costs increase (starting from the initial costs estimates if possible)?*
 - 2.2. *What are the main drivers of the costs increase?*
 - 2.3. *Are there any mitigation measures being discussed or already implemented (e.g. postponement of investments)?*
3. *Would you like to add any comment (general or specific) on this network issue?*

201. This section presents the indicative results of this comparative study, covering a sample of 10 CEER members who answered the questionnaire and/or took part in the bilateral discussions: Italy, France, the Netherlands, Germany, Ireland, Spain, Denmark, Latvia, Estonia and Cyprus.

202. However, the European comparison carried out below presents several limitations that should be highlighted, albeit without affecting the scope of the results obtained. The diversity of the investment projects being compared, the timing of their realisation, the typology of the costs considered and the time horizon in which the cost increase is observed are some of the factors that complicate the study and make the analysis less accurate. However, despite these limitations, the comparison is still relevant in the sense that it makes it possible to identify important general trends - and also differences - and to identify good practices that can inspire reforms.

7.2. EUROPEAN COMPARISON: INDICATIVE RESULTS

203. The results of the survey show that there is also an increase in investment costs in grid infrastructure in the other European countries surveyed (with the exception of Estonia, where projects are still in the planning phase), for both offshore and onshore grid assets. However, the scale of these increases varies from case to case, as illustrated in Table 17. It is a multiplier within a range of 1.03 to 2, and should be seen in the context of specific national characteristics in each case.

Table 17 - Investment costs increase estimates by country

	<i>Costs increase estimate</i>	<i>Rough estimate of multiplier factor based on national realities</i>
Italy (ARERA)	<i>For typical HVDC solution (two converter stations and two cables), compared to 2021, it is roughly a doubling of costs due to a “up to 3x” effect about the costs of the converter stations and “less than 2x” effect about the costs of the DC cables.</i>	x2
France (CRE)	<i>confidential</i>	

Netherlands (ACM)	<i>With regards to the offshore sector, the estimate for the average annual total cost for the period 2032-2057 for the offshore grid necessary to connect 21 GW offshore wind to the onshore network has increased from €2 billion a year in 2022 to €3.6 billion a year in 2023 (see letter to the parliament, p. 6, in Dutch⁹⁹). The estimate of 2022 is not the initial estimate as far as we know. Estimated investment expenditure has increased from €26 billion to €35.5 billion. Onshore, large increases of cost estimates have also been reported by the TSO in the last year.</i>	x1.8
		x1.4 (capex)
Germany (BNetzA)	<i>For several projects, the transmission system operators (TSOs) have also reported cost increases to the BNetzA when awarding contracts for offshore connection lines.</i>	n/a
Latvia (PUC)	<i>Similar problems were especially evident in previous years in PCI projects. In Latvia, there are no offshore sector projects yet, their implementation is at the initial stage of evaluation. Costs increase depends on many factors, for example in the last years one of PCI project investments cost deviation was more than 10% compared to planned costs and it was related to situation in whole world, because demand for specific infrastructure elements was so high, that the price almost doubled compared to initially planned. Most of the cases investment costs increase within 3%, of course no for all projects. For many projects all costs are fixed in the procurement process so deviation in normal situation is not possible.</i>	x1.03
Denmark (DUR)	<i>Regarding the Energy Island Bornholm, the increase in budget is about 30 pct., from 17 bill DKK to 21 bill DKK for the Danish part (leg) of the cables, converters, etc. The budget may be updated soon and could show another change than the 30 pct. These latest budget figures remain very uncertain as they were projected some time ago before the high inflation and problems in the supply chain materialized</i>	x1.3
Estonia (ECA)	<i>The offshore grid project is still in its early stages and the construction procurement process has not yet been initiated. Once the procurement process begins, ECA will be better positioned to address questions related to point 2/if yes. The initial estimated cost range is between 1.2-2 billion euros, depending on which option is chosen as the final location. Currently this is the only available information in Estonia regarding offshore wind park grid.</i>	n/a (project not yet mature)

⁹⁹ Source: [Structuurvisie Windenergie op Zee \(SV WoZ\) | Second Chamber of the States-General](#)

Spain (CNMC)	There is currently an interconnector under construction between Spain and France. For this project, the initial estimated investment costs were 1.750 million euros. In 2020, the project promoters RTE and REE, TSOs respectively in France and Spain, informed about technical changes that increased the investment value of the project up to 1,950 million euros. In 2023, both TSOs informed their respective regulatory authorities of a significant increase in the estimated costs of the project following the preliminary results of the calls for tenders conducted for the purchase of supplies needed to build the interconnector. They asked NRAs to confirm that the costs would be recovered acknowledging the new assessment of the project's costs of €2,850 million, to which is added a provision for risks of €250 million. These changes in costs, according to TSOs, were mainly due to a particularly stressed context on the markets for the supply of HDVC cables and converter stations.	x1.8
Cyprus (CERA)	The total cost for the project connecting Cyprus to Greece is currently estimated at €1.9bn. There was one increase from the initial estimate of €1.5bn to €1.9bn in 2023.	x1.3
Ireland (CRU)	The Celtic Interconnector cable will connect Ireland and France and is expected to be in service by the end of 2026. In April 2019, the NRAs adopted a coordinated decision on the cross-border allocation of the Celtic Interconnector project's investment costs between Ireland and France, with an estimated investment cost of the project being €930 million. The tendering process for the award of the main supply contracts was completed in September 2022, before the energy crisis occurred. The updated estimated cost for the delivery of the Celtic Interconnector (including contingency) is forecast at €1.623 billion. Final overrun costs will be known once the project is delivered and operational.	x1.7

204. Based on the obtained information, this increase can be attributed to several factors, the relative importance of which varies. Reasons or drivers indicated by respondents include:

- inflation (indexation of contracts);
- the increase in raw material costs;
- the increase in labour costs (construction work);
- tensions in the supply chain (in particular due to the limited number of suppliers and the increasing difficulty in finding suppliers who can deliver for specific deadlines);
- the rise in interest rates in the capital market;
- exchange rate fluctuations;
- energy price fluctuations.

205. While tensions in the supply chain, increased commodity prices and inflation are the main drivers of the overall increase in investment costs, the fact remains that in some cases (e.g. Ireland and the Netherlands) this increase is also amplified by initial underestimates of the investment costs (CAPEX and/or OPEX).

206. Indeed, there is a growing general awareness of cost increases and their potential impact on grid tariffs at the European level.

207. To minimise the impact of this increase, or even the escalation of investment costs (Ireland, the Netherlands), risk mitigation measures have been considered, discussed or undertaken in some countries by the relevant public authorities. These include the following:

- updating the underlying analyses of investment projects to verify their effectiveness in a context of higher costs;
- a reassessment of the investment needs (Italy and Spain);
- prioritisation of investments (Italy);
- a postponement or even cancellation of investment projects;

For instance, while the project to build the Bornholm energy island in Denmark is still ongoing, the North Sea Energy Island project has been put on hold indefinitely. It is even possible that this project will never happen. Market tensions may increase or decrease by then, but the final decision lies with the decision-making authorities.

In Italy, specific attention is given to postponing or even cancelling less favourable investment projects, although these projects may have potential benefits that outweigh the costs.

- the use of congestion rents - as envisaged in Spain in the context of its interconnector project with France - to deal with the increase in project costs and absorb them based on the financial stability of the electricity system, in accordance with the provisions of Article 19 of Regulation (EU) No 943/2019 on the internal electricity market.¹⁰⁰ Congestion rents will be used to finance the difference between the CAPEX approved in the latest National Development Plan 2021-2026 and the most recent CAPEX figure.

¹⁰⁰ Source: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019R0943>

- carrying out analyses on the financing of electricity infrastructure; this is the case in the Netherlands in particular, where the government has commissioned a specific study¹⁰¹ by a group of experts, with the aim of gaining more insight and monitoring the scale of investments in electricity infrastructure by 2040 as well as financing for it, from both a national and European perspective. This study should also result in concrete policy options that help improve the country's competitiveness and financing capacity.

The underlying research questions of this analysis (the final report of which is expected in February) are:

- 1) What is the scale of investment needed for the transition up to 2040?
- 2) What are the scenarios for the scale and associated investment costs towards 2040? Effectiveness
- 3) What possibilities are there to increase the efficiency of investments by system operators in the Netherlands and international components or to reduce the social costs?
- 4) How are feasibility, costs and funding of grid costs (incl. offshore grid) included in decision making (both supply and demand stimulation)? Distribution effects with the current structure
- 5) What other costs are passed on in grid tariffs besides investment costs (e.g. grid balancing costs)? And can these costs be influenced?
- 6) Based on what assumptions was the current distribution calculated (e.g. cost reflectivity principle)?
- 7) Where are the costs of expanding and reinforcing the grids borne under the current system?
- 8) What effect do these investments have on the grid tariffs for citizens? (What are different consumer types and what is the impact per type?)
- 9) What effect do these investments have on the grid tariffs for companies? (What are different consumer types and what is the impact for each type, including on competitive conditions?)
- 10) What effect do these investments have on the public finances (balance and debt) in the short and long term?
- 11) What (desired and undesired) effects and risks are associated with the current distribution system? Interventions

¹⁰¹ Source: [BIJLAGE 12: TAAKOPDRACHT INTERDEPARTEMENTAAL BELEIDSONDERZOEK | Ministry of Finance - Government of the Netherlands](#)

- 12) What options are possible to reorganise electricity infrastructure, while taking into account the public interest associated with energy infrastructures such as security of supply and sustainability, in order to reduce the cost to society and budget for electricity infrastructure:
- 1) Supply and demand. What targeted measures can be taken on the electricity supply and demand side to increase social benefits from investments and reduce investment costs?
 - 2) Funding. What policy options are there to distribute investment and other costs of grid operators (electricity) in a different way? (examine broadly, from budget to adaptation)
 - 3) Regulation. Is the legal framework (European and Dutch) and the corresponding competence of ACM still adequate at this stage of the energy transition?
 - 4) Funding. What options are there to redesign the financing structure of system operators in order to limit the costs to society and the budget for electricity infrastructure? In this respect, any implications for governance and the application of the shareholding policy will be outlined in the pros and cons of the options.
 - 5) Preconditions. Which interventions focused on preconditions (e.g. capacity of system operators, spatial planning/energy planning, role of local authorities) are promising to reduce social costs?

208. Besides these risk mitigation measures, we should also mention the recommendation of the Dutch regulator ACM. The aim is to ensure that the government, and more generally the public, receives as much relevant and reliable information as possible from the system operator regarding cost estimates, risks, interdependencies and possible consequences of large-scale infrastructure investments. There is currently no guarantee that the system operator will do this and share all relevant and reliable information at its disposal. However, this information is crucial to ensure that informed decisions are taken regarding the future of the country's energy system. Constructive discussions are currently ongoing with the minister and the grid operator to strengthen the legal framework - in particular the additional powers and roles the regulator should have in this context - in order to improve the overall framework for controlling and monitoring investment projects in electricity grid infrastructure.

209. The Irish regulator, CER, fully agrees with this and urges that:

- data on the economic analysis underpinning major projects and policy choices are transparently communicated between government departments involved in the energy transition;
- after the initial analysis, the relevant data related to a cost increase are shared transparently and regularly, so that there is continuous awareness of the cost increase (when this occurs) and the possible consequences.

Finally, given the fact that Europe is highly dependent on a limited number of foreign suppliers, namely for converter stations, additional initiatives are proposed at the European level within CEER. In this respect, further discussions about supply chain tensions in the context of the debate on European competitiveness, as well as a follow-up on future challenges (identifying needs and prioritising projects) would help provide relevant insights.

8. RECOMMENDATIONS OF THE CREG REGARDING THE LEGAL FRAMEWORK

210. Given the impact of the federal development plan on grid tariffs, it is crucial to enhance the transparency and reliability of the approval process to ensure that it is based on solid and verifiable foundations. In this context, the obligations of the transmission system operator (TSO) should be strengthened, so that they present a figures-based plan with a detailed cost-benefit analyses for each major investment project (including the different options and variants). The CREG also recommends introducing an alert mechanism, by requiring that the TSO provides a detailed report in the event of a significant increase in investment costs for a major project, or a significant overrun of the planned implementation deadlines, so that the CREG can propose changes to the development plan where appropriate.

The CREG also asks to be explicitly involved at each stage of the drafting of the development plan, to help define the plan's objectives and priorities from the outset. To this end, the CREG should be able to verify the identification of needs and the cost-benefit analyses, with guaranteed access to all data deemed relevant. This enhanced role will allow the CREG to ensure the adequacy and transparency of investments in the grid more effectively, to meet the needs of the energy transition while protecting the interests of consumers.

The CREG asks that its collaboration with the General Directorate of Energy and the Federal Planning Bureau be formalised. A collaboration and monitoring committee made up of these three bodies should be set up for the development plan. In order to have a mechanism to exert influence prior to the adoption of the federal development plan, the CREG also requests that it is able to issue a unified, i.e. binding, opinion on the development plan and that the deadline for issuing this opinion be extended to at least 60 days, instead of the current 40 days, to allow it to make more in-depth analyses.

Furthermore, the CREG recommends including in Article 12, § 5 of the Electricity Act a guideline on tariffs that encourages the system operator to improve its budget estimates for major investment projects and foster proactive management to avoid budget overruns that are inadequately justified by exceptional circumstances. Such overruns may have consequences on the recovery of their costs and/or the corresponding compensation through tariffs.

The impact of the cost increase due to the MOG II will primarily be felt in the subsequent tariff periods. When the new tariff methodology for the 2028-2032 period will be drafted, the CREG will examine possible alternatives to the existing regulatory model - where the remuneration for the system operator is largely determined by the scale of the investments - to encourage the latter to optimise the use of existing assets and minimise investment costs.

Finally, Article 26, §§2 and 3 of the Electricity Act should be revised to prevent professional secrecy from being invoked when the CREG is required to communicate confidential information to the Chamber and to the Minister for Energy, on condition that the authority that will receive this information undertakes to ensure the same level of confidentiality as that required of the CREG.

9. CONCLUSION

9.1. GENERAL CONCLUSIONS

In view of the foregoing analyses, the following can be stated:

- the CREG, despite its multidisciplinary expertise, currently has too limited legal powers in the context of the drafting, approval and possible revision of network development plans and in the context of monitoring infrastructure projects developed by the TSO;
- the CREG faced numerous difficulties in the context of monitoring cost evolution in connection with the MOG II dossier. These include in particular:
 - the TSO's lack of transparency regarding all the information which is useful for the proper exercise of this oversight;
 - the lack of quantitative and qualitative analyses that were essential before the federal government approved the design of the MOG II in 2021;
 - the TSO's refusal to analyse alternative options for the design of the MOG II at the request of the CREG in June 2024, to which there was no positive follow-up until December 2024, in order to weigh up the alternatives and make informed investment choices in the public interest and in the interest of Belgian consumers;
 - Up until October 2024, the lack of a proactive and timely communication by the TSO to the competent authorities regarding important developments in the case, including the identified budget overrun and the unilateral decisions of the TSO relating in particular to significant changes in the choice of technical options that led to the budget increases.
- The budget for the MOG II was revised upwards, from € 2.2 billion in 2021 to [CONFIDENTIAL: between € 7 and 8 billion] currently, according to Elia's latest estimates.

If the current design and budget of the MOG II remain unchanged, the cost, to be covered by transmission tariffs, is expected to be around € 800 million per year from 2034, when the full infrastructure will be commissioned.

These costs correspond to € 9.8/MWh (about € 34.4/year) for a residential consumer and € 4.45/MWh for a large industrial consumer directly connected to the Elia 380/220/150 kV transmission grid.

- The CREG has analysed the budget increases of the current MOG II design.
 - The CREG concludes that the budget increases of the MOG II project can be attributed to:
 - the market developments for HVDC and AC infrastructure over the period 2022-2024, due to global scarcity in the supply chain, and in particular for offshore applications. These market developments have resulted in limited supply, with higher bids, less flexibility and more risk for Elia;
 - Elia's focus on ensuring a grid design with high availability and operational flexibility led to assumptions and design choices that increased the cost base compared to typical offshore transmission designs. Examples of this include operation with a single AC node, the higher capacity of the HVDC connection and including too high redundancy and too much spare equipment;
 - adherence to strict timing for delivery, whereby unverified assumptions have resulted in premature design choices. The main factors in this regard are the following: the assumption that the offshore HVDC and AC substations could be built on the island through onshore techniques (stick-built method) and the technical and regulatory feasibility of using a single AC node.
 - based on the detailed analysis carried out by the CREG and DNV, cost-increasing effects of at least € 1.570 billion have been identified that can be directly linked to specific (technical) choices made by Elia.
 - the wish of the transmission system operator (TSO) to develop a first-of-a-kind infrastructure, supported by a government decision in the context of the European Recovery Plan, the grant for which is currently uncertain and the amount limited;
- In addition to these aspects, a number of possible additional budget increases were identified that could further (significantly) increase the above-mentioned amount.
- Based on the alternative options and other information communicated by Elia on 13 December 2024 and 8 January 2025, option 3 (Nautilus point-to-point and 2.1 GW of offshore wind in the PEZ) appears *a priori* - subject to the working assumptions applied for the modelling and subsequent analyses - more favourable from the Belgian consumers' perspective. While this option has significant potential to reduce investment costs, it should also be seen in light of the uncertainties that currently impact the Nautilus project as a whole. Moreover, any potential decision that gives preference to alternative options that do not envisage DC installations on the island would also mean looking again at the optimal size of the island and the design of the electricity system, which could lead to additional savings.

- At the European level, a general rising trend in investment costs can also be seen in the context of large infrastructure projects, including offshore grid assets. While tensions in the supply chain, higher commodity prices and inflation are the main drivers for this general rise in investment costs, the fact remains that such rises can also be exacerbated by initial underestimates of investment costs by the system operator in question. To minimise the impact of this rise, or even escalation, of the investment costs, risk mitigation measures are also being considered and discussed by the different countries investigated.

9.2. RECOMMENDATIONS

The recommendations below aim to improve the regulatory framework for investment projects in grid infrastructure. They pertain both to the MOG II project examined here and to investment projects that are currently ongoing or future projects.

First, it appears crucial that grid infrastructure projects, as they address issues of public interest, be subject to legal procedures that guarantee, in particular, the objectivity and transparency of all qualitative and quantitative analyses underpinning them. This requirement is essential to ensure that the results of these analyses only serve the public interest, without being influenced by potential conflicts of interest or other specific considerations.

To ensure that the public interest remains the absolute priority in guiding the decisive investment choices for the energy future of Belgium, the Electricity Act should be amended with regard to:

- the current procedure for preparing, approving and following-up grid development plans, in particular by:
 - strengthening the role of the public authorities, including that of the CREG;
 - obliging the system operator to present, at the very early stages of the development plan, all possible investment options, backed up by quantified analyses to meet the same grid need, so that the authorities have a certain decision-making flexibility, to then adequately describe the conceptual design in the development plan, to avoid situations where the competent and decision-making authorities are in a bind when alternative choices and courses of action are restricted or even non-existent, to the detriment of the public interest;
 - a strict mechanism for budgetary oversight of the investment projects of the transmission system operator, including periodic reports to the competent and decision-making authorities, describing the evolution of costs compared to initial estimates, as well as the explanatory factors for any identified discrepancies, so that any budget overruns or even escalations are identified, and corrective measures can be assessed in good time.
- the tariff framework to ensure better control of investment costs and foster proactive management on the part of the transmission system operator in the interest of grid users through a new provision in the tariff guidelines to provide tangible impetus for strict budgetary oversight;

- the restrictions relating to professional secrecy currently imposed on the CREG, to allow genuine parliamentary debate and democratic oversight on issues of public interest and on the services provided by the CREG, especially at the request of the government.

In addition to these considerations, there is also a need to take into account the possible views of different stakeholders as much as possible throughout the process, based on public consultations, so as to consolidate public support for decisive investment projects in the context of the energy transition.

The CREG wishes to emphasise that, in the context of the drafting of this report, from October 2024 onwards, collaboration with the TSO has proved constructive and recommends that openness and transparency towards the CREG be maintained.

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